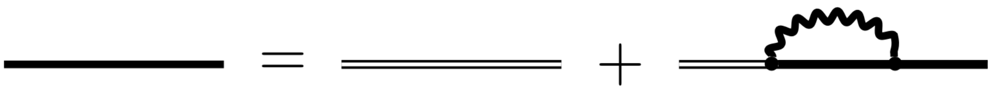
This is a part of SFQED-Loops script collection
developed for calculating loop and tree-level processes in
Strong-Field Quantum Electrodynamics.
The scripts are available on https://github.com/ArsenyMironov/SFQED-Loops

If you use this script in your research, please, consider
citing our papers:
- A. A. Mironov, S. Meuren, and A. M. Fedotov, PRD 102, 053005 (2020); arXiv:2003.06909
  https://doi.org/10.1103/PhysRevD.102.053005
- A. A. Mironov, A. M. Fedotov, 105 (3), 033005 (2022); arXiv:2109.00634
https://doi.org/10.1103/PhysRevD.105.033005
- Y. V. Selivanov, A. A. Mironov, A. I. Alexeenko, A. M. Fedotov arXiv:2605.12479 (2026)

If you have any questions, please, don’t hesitate to contact:
mironov.hep@gmail.com



In[1]:=
```
NotebookEvaluate[NotebookOpen[NotebookDirectory[] <> "definitions.nb"]]
```

FeynCalc 10.1.0 (stable version). For help, use the online documentation, visit the forum and have a look at the supplied examples. The PDF-version of the manual can be downloaded here.

If you use FeynCalc in your research, please evaluate FeynCalcHowToCite[] to learn how to cite this software.

Please keep in mind that the proper academic attribution of our work is crucial to ensure the future development of this package!

Electron propagator in $E_p - \mathtt{representation}$ satisfies the equation
$$S^c(p, q) = (2\pi)^4 \delta(p - q) S^c(p)$$

$$[\gamma p - m - M(p)]\, S^c(p) = i$$

Let

$$D(p) = \gamma p - m - M(p) = S + \gamma V + \sigma T + \gamma A \gamma^5$$

$$S = m\, s\left(p^2, \chi_p\right),$$

$$V^\mu = p^\mu v_1\left(p^2, \chi_p\right) + \frac{e^2 \left(F^2 p\right)^\mu}{m^4 \chi^2} v_2\left(p^2, \chi_p\right),$$

$$T_{\mu\nu} = \frac{e F_{\mu\nu}}{m \chi} t\left(p^2, \chi_p\right),$$

$$A^\mu = \frac{e\,(F^* p)^\mu}{m^2 \chi} a_s\left(p^2, \chi_p\right),$$ where s stands for ' scalar';

then

$$S^c(p) = i D^{-1}(p) = \frac{i}{2}\left(S - \gamma V - \sigma T + \gamma A \gamma^5\right)\left[\frac{1}{D_+\left(p^2, \chi_p\right)}\left(1 + \gamma\epsilon^{(2)} \gamma^5\right) + \frac{1}{D_-\left(p^2, \chi_p\right)}\left(1 - \gamma\epsilon^{(2)} \gamma^5\right)\right] =$$

$$= \frac{i}{2}\left[m\, s\left(p^2, \chi_p\right) - (\gamma p)\, v_1\left(p^2, \chi_p\right) - \frac{e^2 \left(\gamma F^2 p\right)}{m^4} v_2\left(p^2, \chi_p\right) - \frac{e\, \sigma F}{m} t\left(p^2, \chi_p\right) + \frac{e\,(\gamma F^* p)\, \gamma^5}{m^2} a_s\left(p^2, \chi_p\right)\right] \times$$

$$\left[\frac{1}{D_+\left(p^2, \chi_p\right)}\left(1 + \gamma\epsilon^{(2)} \gamma^5\right) + \frac{1}{D_-\left(p^2, \chi_p\right)}\left(1 - \gamma\epsilon^{(2)} \gamma^5\right)\right],$$

$$\epsilon^{(2)}{}_\mu = \frac{e\,(F^* p)^\mu}{m^3 \chi_p},$$

Electron propagator

$$S^c(x_2, x_1) = \Lambda^{4-D} \int \frac{d^D p}{(2\pi)^D} E_p(x_2)\, S^c(p)\, \overline{E}_p(x_1)$$

$$x = x_2 - x_1,$$

$$X = \frac{1}{2}(x_1 + x_2),$$

$$\xi^2 = -\frac{e^2 a^2}{m^2},$$

$[\Lambda] = m$ – mass scale,

$$E_p\ (x_2)\ =\ \left[1-\frac{e\ (\gamma k)\ (\gamma a)}{2\ (kp)}\ (kx_2)\right]$$
$$\text{Exp}\left[-i\ (p\ x_2)\ +\ i\ \frac{e\ (a\ p)}{2\ (k\ p)}\ (k\ x_2)^2+i\ \frac{a^2\ e^2}{6\ (k\ p)}\ (k\ x_2)^3\right];$$

In[2]:=
```
NewMomentum["p"]
NewCoordinate["x1"]
NewCoordinate["x2"]
NewCoordinate["x"]
NewCoordinate["X"]
```

$$\left\{p^\alpha, p^2, k\cdot p, \text{Fp}^\alpha, \text{FFp}^\alpha, \text{FDp}^\alpha, a\cdot p, 0, 0, 0, -a^2\,(k\cdot p), 0, 0, -\frac{m^6\,\chi\text{p}^2}{e^2}, -\frac{m^6\,\chi\text{p}^2}{e^2}, \frac{m^6\,\chi\text{p}^2}{e^2}, 0, 0, 0, 0, 0, 0\right\}$$

$$\left\{\text{x1}^\alpha, \text{x1}^2, k\cdot\text{x1}, a\cdot\text{x1}, \text{Fx1}^\alpha, \text{FFx1}^\alpha, \text{FDx1}^\alpha, k\cdot\text{x1}, 0, 0, 0, -a^2\,(k\cdot\text{x1}), 0, 0, -\frac{m^2\,\xi^2\,(k\cdot\text{x1})^2}{e^2}, -\frac{m^2\,\xi^2\,(k\cdot\text{x1})^2}{e^2}, \frac{m^2\,\xi^2\,(k\cdot\text{x1})^2}{e^2}, 0, 0, 0, 0, 0, 0\right\}$$

$$\left\{\text{x2}^\alpha, \text{x2}^2, k\cdot\text{x2}, a\cdot\text{x2}, \text{Fx2}^\alpha, \text{FFx2}^\alpha, \text{FDx2}^\alpha, k\cdot\text{x2}, 0, 0, 0, -a^2\,(k\cdot\text{x2}), 0, 0, -\frac{m^2\,\xi^2\,(k\cdot\text{x2})^2}{e^2}, -\frac{m^2\,\xi^2\,(k\cdot\text{x2})^2}{e^2}, \frac{m^2\,\xi^2\,(k\cdot\text{x2})^2}{e^2}, 0, 0, 0, 0, 0, 0\right\}$$

$$\left\{x^\alpha, x^2, k\cdot x, a\cdot x, \text{Fx}^\alpha, \text{FFx}^\alpha, \text{FDx}^\alpha, k\cdot x, 0, 0, 0, -a^2\,(k\cdot x), 0, 0, -\frac{m^2\,\xi^2\,(k\cdot x)^2}{e^2}, -\frac{m^2\,\xi^2\,(k\cdot x)^2}{e^2}, \frac{m^2\,\xi^2\,(k\cdot x)^2}{e^2}, 0, 0, 0, 0, 0, 0\right\}$$

$$\left\{X^\alpha, X^2, k\cdot X, a\cdot X, \text{FX}^\alpha, \text{FFX}^\alpha, \text{FDX}^\alpha, k\cdot X, 0, 0, 0, -a^2\,(k\cdot X), 0, 0, -\frac{m^2\,\xi^2\,(k\cdot X)^2}{e^2}, -\frac{m^2\,\xi^2\,(k\cdot X)^2}{e^2}, \frac{m^2\,\xi^2\,(k\cdot X)^2}{e^2}, 0, 0, 0, 0, 0, 0\right\}$$

$$\text{DdInv}[\text{sgn}] == D_{\text{sgn}}\left(p^2, \chi_p\right)$$
$$\in\ ==\ \in^{(2)}$$

In[7]:=
```
FToak = {Ft[α_, β_] → kv[α] × av[β] - av[α] × kv[β], DiracGamma[Momentum[Fp, D], D] →
    DiracGamma[Momentum[k, D], D] × Pair[Momentum[a, D], Momentum[p, D]] -
     DiracGamma[Momentum[a, D], D] × Pair[Momentum[k, D], Momentum[p, D]],
   DiracGamma[Momentum[FFp, D], D] →
    -av2 DiracGamma[Momentum[k, D], D] × Pair[Momentum[k, D], Momentum[p, D]],
   FFpv[μ_] → -av2 kp kv[μ]}
FToEps = {FDpv[μ_] →
    Contract[1 / 2 Eps[LorentzIndex[μ, D], LorentzIndex[ν, D], LorentzIndex[α2, D],
       LorentzIndex[α3, D]] (kv[α2] × av[α3] - av[α2] × kv[α3]) pv[ν]]}
Gamma5toTrippleGamma =
  {GAD[μ_].GA[5] × Eps[LorentzIndex[μ_, D], Momentum[a_, D], Momentum[b_, D],
      Momentum[c_, D]] → Contract[I Pair[LorentzIndex[α1, D], Momentum[a, D]] ×
      Pair[LorentzIndex[α2, D], Momentum[b, D]] ×
      Pair[LorentzIndex[α3, D], Momentum[c, D]] (GAD[α1, α2, α3] -
        ( MTD[α1, α2] × GAD[α3] + MTD[α2, α3] × GAD[α1] - MTD[α1, α3] × GAD[α2]))]}
```

Out[7]= $\{F(\alpha\_, \beta\_) \to a^{\beta} k^{\alpha} - a^{\alpha} k^{\beta}, \gamma\cdot\mathrm{Fp} \to (a\cdot p)\,\gamma\cdot k - \gamma\cdot a\,(k\cdot p), \gamma\cdot\mathrm{FFp} \to a^2\,(-(\gamma\cdot k))\,(k\cdot p), \mathrm{FFp}^{\mu\_} \to a^2\,(-k^{\mu})\,(k\cdot p)\}$

Out[8]= $\{\mathrm{FDp}^{\mu\_} \to -\epsilon^{\mu a k p}\}$

Out[9]= $\{\gamma^{\mu\_}.\bar{\gamma}^5\,\epsilon^{\mu\_ \mathrm{a\_ b\_ c\_}} \to i\,(-(a\cdot b)\,\gamma\cdot c + (a\cdot c)\,\gamma\cdot b - \gamma\cdot a\,(b\cdot c) + (\gamma\cdot a).(\gamma\cdot b).(\gamma\cdot c))\}$

In[10]:=
```
FxToak =
  {Ft[α_, β_] → kv[α] × av[β] - av[α] × kv[β], DiracGamma[Momentum[Fx, D], D] →
    DiracGamma[Momentum[k, D], D] × Pair[Momentum[a, D], Momentum[x, D]] -
     DiracGamma[Momentum[a, D], D] × Pair[Momentum[k, D], Momentum[x, D]],
   DiracGamma[Momentum[FFx, D], D] →
    -av2 DiracGamma[Momentum[k, D], D] × Pair[Momentum[k, D], Momentum[x, D]],
   FFxv[μ_] → -av2 kx kv[μ],
   σF → -2 I DiracGamma[Momentum[a, D], D].DiracGamma[Momentum[k, D], D] }
FxToEps = {FDx[μ_] →
    Contract[1 / 2 Eps[LorentzIndex[μ, D], LorentzIndex[ν, D], LorentzIndex[α2, D],
       LorentzIndex[α3, D]] (kv[α2] × av[α3] - av[α2] × kv[α3]) xv[ν]],
   DiracGamma[Momentum[FDx, D], D].DiracGamma[5] →
    Contract[1 / 2 Eps[LorentzIndex[μ, D], LorentzIndex[ν, D],
       LorentzIndex[α2, D], LorentzIndex[α3, D]]
      (kv[α2] × av[α3] - av[α2] × kv[α3]) xv[ν] × GAD[μ].DiracGamma[5]]}
Gamma5toTrippleGammax =
  {DiracGamma[LorentzIndex[μ_, D], D].DiracGamma[5] Eps[LorentzIndex[μ_, D],
      Momentum[a_, D], Momentum[b_, D], Momentum[c_, D]] → Contract[
     I Pair[LorentzIndex[α1, D], Momentum[a, D]] × Pair[LorentzIndex[α2, D],
       Momentum[b, D]] × Pair[LorentzIndex[α3, D], Momentum[c, D]] (GAD[α1, α2, α3] -
        ( MTD[α1, α2] × GAD[α3] + MTD[α2, α3] × GAD[α1] - MTD[α1, α3] × GAD[α2]))]}
```

Out[10]= $\{F(\alpha\_, \beta\_) \to a^{\beta} k^{\alpha} - a^{\alpha} k^{\beta}, \gamma\cdot\mathrm{Fx} \to (a\cdot x)\,\gamma\cdot k - \gamma\cdot a\,(k\cdot x),$
$\gamma\cdot\mathrm{FFx} \to a^2\,(-(\gamma\cdot k))\,(k\cdot x), \mathrm{FFx}^{\mu\_} \to a^2\,(-k^{\mu})\,(k\cdot x), \sigma\mathrm{F} \to -2\,i\,(\gamma\cdot a).(\gamma\cdot k)\}$

Out[11]= $\{\mathrm{FDx}(\mu\_) \to -\epsilon^{\mu a k x}, (\gamma\cdot\mathrm{FDx}).\bar{\gamma}^5 \to -\gamma^{\mu}.\bar{\gamma}^5\,\epsilon^{\mu a k x}\}$

Out[12]= $\{\gamma^{\mu\_}.\bar{\gamma}^5\,\epsilon^{\mu\_ \mathrm{a\_ b\_ c\_}} \to i\,(-(a\cdot b)\,\gamma\cdot c + (a\cdot c)\,\gamma\cdot b - \gamma\cdot a\,(b\cdot c) + (\gamma\cdot a).(\gamma\cdot b).(\gamma\cdot c))\}$

In[13]:=
```
S = m sf
V[μ_] = pv[μ] v1 + e^2 FFpv[μ] / m^4 / χp^2 v2
T[μ_, ν_] = e Ft[μ, ν] / m / χp tf
A[μ_] = e FDpv[μ] / m^2 / χp af
∈[μ_] = e FDpv[μ] / m^3 / χp
```

Out[13]= $m\,\mathrm{sf}$

Out[14]= $$\frac{e^2\,\mathrm{v2}\,\mathrm{FFp}^{\mu}}{m^4\,\chi\mathrm{p}^2}+\mathrm{v1}\,p^{\mu}$$

Out[15]= $$\frac{e\,\mathrm{tf}\,F(\mu,\nu)}{m\,\chi\mathrm{p}}$$

Out[16]= $$\frac{\mathrm{af}\,e\,\mathrm{FDp}^{\mu}}{m^2\,\chi\mathrm{p}}$$

Out[17]= $$\frac{e\,\mathrm{FDp}^{\mu}}{m^3\,\chi\mathrm{p}}$$

In[18]:=
```
Scp[sgn_] =
  (S - GAD[α] × V[α] - I / 2 (GAD[α, β] - GAD[β, α]) T[α, β] + A[α] × GAD[α].GA[5]).
      (1 + sgn ∈[μ] GAD[μ].GA[5]) /. FToEps /. FToak /. Gamma5toTrippleGamma
```

Out[18]= $$\left(-\frac{i\,\mathrm{af}\,e\,((a\cdot p)\,\gamma\cdot k-\gamma\cdot a\,(k\cdot p)+(\gamma\cdot a).(\gamma\cdot k).(\gamma\cdot p))}{m^2\,\chi\mathrm{p}}-\gamma^{\alpha}\left(\mathrm{v1}\,p^{\alpha}-\frac{a^2\,e^2\,\mathrm{v2}\,k^{\alpha}\,(k\cdot p)}{m^4\,\chi\mathrm{p}^2}\right)-\frac{i\,e\,\mathrm{tf}\left(\gamma^{\alpha}.\gamma^{\beta}-\gamma^{\beta}.\gamma^{\alpha}\right)\left(a^{\beta}\,k^{\alpha}-a^{\alpha}\,k^{\beta}\right)}{2\,m\,\chi\mathrm{p}}+m\,\mathrm{sf}\right).\left(1-\frac{i\,e\,\mathrm{sgn}\,((a\cdot p)\,\gamma\cdot k-\gamma\cdot a\,(k\cdot p)+(\gamma\cdot a).(\gamma\cdot k).(\gamma\cdot p))}{m^3\,\chi\mathrm{p}}\right)$$

In[19]:=
```
Epx2 = Ep[x2, p]
EpBarx1 = EpC[x1, p]
```

Out[19]= $$\left\{1-\frac{e\,(k\cdot\mathrm{x2})\,(\gamma\cdot k).(\gamma\cdot a)}{2\,(k\cdot p)},\ \frac{a^2\,e^2\,(k\cdot\mathrm{x2})^3}{6\,(k\cdot p)}+\frac{e\,(a\cdot p)\,(k\cdot\mathrm{x2})^2}{2\,(k\cdot p)}-p\cdot\mathrm{x2}\right\}$$

Out[20]= $$\left\{1-\frac{e\,(k\cdot\mathrm{x1})\,(\gamma\cdot a).(\gamma\cdot k)}{2\,(k\cdot p)},\ -\frac{a^2\,e^2\,(k\cdot\mathrm{x1})^3}{6\,(k\cdot p)}-\frac{e\,(a\cdot p)\,(k\cdot\mathrm{x1})^2}{2\,(k\cdot p)}+p\cdot\mathrm{x1}\right\}$$

```
In[21]:= Matrix = Epx2[[1]].Scp[ζ].EpBarx1[[1]]
Coeff = i / 2 Λ^(4 - D) / (2 π)^D / DdInv[ζ]
Phase = Epx2[[2]] + EpBarx1[[2]]
```

Out[21]=
$$\left(1-\frac{e\,(k\cdot \mathrm{x2})\,(\gamma\cdot k).(\gamma\cdot a)}{2\,(k\cdot p)}\right).\left(-\frac{i\,\mathrm{af}\,e\,((a\cdot p)\,\gamma\cdot k-\gamma\cdot a\,(k\cdot p)+(\gamma\cdot a).(\gamma\cdot k).(\gamma\cdot p))}{m^2\,\chi\mathrm{p}}-\gamma^{\alpha}\left(\mathrm{v1}\,p^{\alpha}-\frac{a^2\,e^2\,\mathrm{v2}\,k^{\alpha}\,(k\cdot p)}{m^4\,\chi\mathrm{p}^2}\right)-\frac{i\,e\,\mathrm{tf}\left(\gamma^{\alpha}.\gamma^{\beta}-\gamma^{\beta}.\gamma^{\alpha}\right)\left(a^{\beta}\,k^{\alpha}-a^{\alpha}\,k^{\beta}\right)}{2\,m\,\chi\mathrm{p}}+m\,\mathrm{sf}\right).\left(1-\frac{i\,e\,\zeta\,((a\cdot p)\,\gamma\cdot k-\gamma\cdot a\,(k\cdot p)+(\gamma\cdot a).(\gamma\cdot k).(\gamma\cdot p))}{m^3\,\chi\mathrm{p}}\right).\left(1-\frac{e\,(k\cdot \mathrm{x1})\,(\gamma\cdot a).(\gamma\cdot k)}{2\,(k\cdot p)}\right)$$

Out[22]=
$$\frac{i\,2^{-D-1}\,\pi^{-D}\,\Lambda^{4-D}}{\mathrm{DdInv}(\zeta)}$$

Out[23]=
$$-\frac{a^2\,e^2\,(k\cdot \mathrm{x1})^3}{6\,(k\cdot p)}+\frac{a^2\,e^2\,(k\cdot \mathrm{x2})^3}{6\,(k\cdot p)}-\frac{e\,(a\cdot p)\,(k\cdot \mathrm{x1})^2}{2\,(k\cdot p)}+\frac{e\,(a\cdot p)\,(k\cdot \mathrm{x2})^2}{2\,(k\cdot p)}+p\cdot \mathrm{x1}-p\cdot \mathrm{x2}$$

```
In[24]:= Matrix1 = Contract[DiracSimplify[Matrix]]
Coeff1 = Coeff;
Phase1 = Phase;
```

Out[24]=
$$\frac{i\,\mathrm{v2}\,\zeta\,(\gamma\cdot k).(\gamma\cdot a)\,a^2\,(k\cdot p)^2\,e^3}{m^7\,\chi\mathrm{p}^3}+\frac{\mathrm{af}\,\zeta\,(\gamma\cdot a).(\gamma\cdot k)\,a^2\,(k\cdot p)\,(k\cdot \mathrm{x1})\,e^3}{2\,m^5\,\chi\mathrm{p}^2}+\frac{\mathrm{af}\,\zeta\,(\gamma\cdot k).(\gamma\cdot a)\,a^2\,(k\cdot p)\,(k\cdot \mathrm{x2})\,e^3}{2\,m^5\,\chi\mathrm{p}^2}-$$
$$\frac{i\,\mathrm{v1}\,\zeta\,(\gamma\cdot a).(\gamma\cdot k)\,a^2\,(k\cdot \mathrm{x1})\,(k\cdot \mathrm{x2})\,e^3}{2\,m^3\,\chi\mathrm{p}}-\frac{\mathrm{af}\,\zeta\,a^2\,(k\cdot p)^2\,e^2}{m^5\,\chi\mathrm{p}^2}+\frac{\mathrm{v2}\,\gamma\cdot k\,a^2\,(k\cdot p)\,e^2}{m^4\,\chi\mathrm{p}^2}+\frac{2\,\mathrm{tf}\,\zeta\,\gamma\cdot k\,a^2\,(k\cdot p)\,e^2}{m^4\,\chi\mathrm{p}^2}-$$
$$\frac{\mathrm{af}\,\zeta\,(\gamma\cdot a).(\gamma\cdot k)\,(a\cdot p)\,(k\cdot p)\,e^2}{m^5\,\chi\mathrm{p}^2}-\frac{\mathrm{af}\,\zeta\,(\gamma\cdot k).(\gamma\cdot a)\,(a\cdot p)\,(k\cdot p)\,e^2}{m^5\,\chi\mathrm{p}^2}+\frac{i\,\mathrm{sf}\,\zeta\,\gamma\cdot k\,a^2\,(k\cdot \mathrm{x1})\,e^2}{2\,m^2\,\chi\mathrm{p}}+\frac{i\,\mathrm{af}\,\gamma\cdot k\,a^2\,(k\cdot \mathrm{x1})\,e^2}{2\,m^2\,\chi\mathrm{p}}-$$
$$\frac{i\,\mathrm{v1}\,\zeta\,(\gamma\cdot p).(\gamma\cdot k)\,a^2\,(k\cdot \mathrm{x1})\,e^2}{2\,m^3\,\chi\mathrm{p}}-\frac{i\,\mathrm{sf}\,\zeta\,\gamma\cdot k\,a^2\,(k\cdot \mathrm{x2})\,e^2}{2\,m^2\,\chi\mathrm{p}}-\frac{i\,\mathrm{af}\,\gamma\cdot k\,a^2\,(k\cdot \mathrm{x2})\,e^2}{2\,m^2\,\chi\mathrm{p}}+\frac{i\,\mathrm{v1}\,\zeta\,(\gamma\cdot k).(\gamma\cdot p)\,a^2\,(k\cdot \mathrm{x2})\,e^2}{2\,m^3\,\chi\mathrm{p}}+$$
$$\frac{i\,\mathrm{v1}\,\zeta\,(\gamma\cdot a).(\gamma\cdot k)\,(a\cdot p)\,(k\cdot \mathrm{x2})\,e^2}{m^3\,\chi\mathrm{p}}+\frac{i\,\mathrm{v1}\,\zeta\,(\gamma\cdot k).(\gamma\cdot a)\,(a\cdot p)\,(k\cdot \mathrm{x2})\,e^2}{m^3\,\chi\mathrm{p}}+\frac{\mathrm{v1}\,\gamma\cdot k\,a^2\,(k\cdot \mathrm{x1})\,(k\cdot \mathrm{x2})\,e^2}{2\,(k\cdot p)}+$$
$$\frac{i\,\mathrm{tf}\,(\gamma\cdot a).(\gamma\cdot k)\,e}{m\,\chi\mathrm{p}}-\frac{i\,\mathrm{tf}\,(\gamma\cdot k).(\gamma\cdot a)\,e}{m\,\chi\mathrm{p}}-\frac{i\,\mathrm{sf}\,\zeta\,(\gamma\cdot a).(\gamma\cdot k).(\gamma\cdot p)\,e}{m^2\,\chi\mathrm{p}}-\frac{i\,\mathrm{af}\,(\gamma\cdot a).(\gamma\cdot k).(\gamma\cdot p)\,e}{m^2\,\chi\mathrm{p}}-\frac{i\,\mathrm{sf}\,\zeta\,\gamma\cdot k\,(a\cdot p)\,e}{m^2\,\chi\mathrm{p}}-$$
$$\frac{i\,\mathrm{af}\,\gamma\cdot k\,(a\cdot p)\,e}{m^2\,\chi\mathrm{p}}+\frac{2\,i\,\mathrm{v1}\,\zeta\,(\gamma\cdot k).(\gamma\cdot p)\,(a\cdot p)\,e}{m^3\,\chi\mathrm{p}}+\frac{i\,\mathrm{v1}\,\zeta\,(\gamma\cdot p).(\gamma\cdot k)\,(a\cdot p)\,e}{m^3\,\chi\mathrm{p}}+\frac{i\,\mathrm{sf}\,\zeta\,\gamma\cdot a\,(k\cdot p)\,e}{m^2\,\chi\mathrm{p}}+\frac{i\,\mathrm{af}\,\gamma\cdot a\,(k\cdot p)\,e}{m^2\,\chi\mathrm{p}}-$$
$$\frac{2\,i\,\mathrm{v1}\,\zeta\,(\gamma\cdot a).(\gamma\cdot p)\,(k\cdot p)\,e}{m^3\,\chi\mathrm{p}}-\frac{i\,\mathrm{v1}\,\zeta\,(\gamma\cdot p).(\gamma\cdot a)\,(k\cdot p)\,e}{m^3\,\chi\mathrm{p}}-\frac{m\,\mathrm{sf}\,(\gamma\cdot a).(\gamma\cdot k)\,(k\cdot \mathrm{x1})\,e}{2\,(k\cdot p)}+\frac{\mathrm{v1}\,(\gamma\cdot p).(\gamma\cdot a).(\gamma\cdot k)\,(k\cdot \mathrm{x1})\,e}{2\,(k\cdot p)}-$$
$$\frac{m\,\mathrm{sf}\,(\gamma\cdot k).(\gamma\cdot a)\,(k\cdot \mathrm{x2})\,e}{2\,(k\cdot p)}+\frac{\mathrm{v1}\,(\gamma\cdot k).(\gamma\cdot a).(\gamma\cdot p)\,(k\cdot \mathrm{x2})\,e}{2\,(k\cdot p)}+\frac{i\,\mathrm{v1}\,\zeta\,(\gamma\cdot a).(\gamma\cdot k)\,p^2\,e}{m^3\,\chi\mathrm{p}}+m\,\mathrm{sf}-\mathrm{v1}\,\gamma\cdot p$$

In[27]:=
```
(*a small test*)
((Matrix1 /. {sf → 1, v1 → -1, v2 → 0, t → 0, af → 0, ξ → 1}) +
   (Matrix1 /. {sf → 1, v1 → -1, v2 → 0, t → 0, af → 0, ξ → -1})) / 2
```

Out[27]=
$$\frac{1}{2}\left(-\frac{a^2 e^2 \gamma\cdot k\,(k\cdot \mathrm{x1})\,(k\cdot \mathrm{x2})}{k\cdot p}-\frac{e\,m\,(k\cdot \mathrm{x1})\,(\gamma\cdot a).(\gamma\cdot k)}{k\cdot p}-\frac{e\,m\,(k\cdot \mathrm{x2})\,(\gamma\cdot k).(\gamma\cdot a)}{k\cdot p}+\frac{2\,i\,e\,\mathrm{tf}\,(\gamma\cdot a).(\gamma\cdot k)}{m\,\chi\mathrm{p}}-\frac{2\,i\,e\,\mathrm{tf}\,(\gamma\cdot k).(\gamma\cdot a)}{m\,\chi\mathrm{p}}-\frac{e\,(k\cdot \mathrm{x1})\,(\gamma\cdot p).(\gamma\cdot a).(\gamma\cdot k)}{k\cdot p}-\frac{e\,(k\cdot \mathrm{x2})\,(\gamma\cdot k).(\gamma\cdot a).(\gamma\cdot p)}{k\cdot p}+2\,m+2\,\gamma\cdot p\right)$$

In[28]:=
```
Matrix2 = Expand[ExpandScalarProduct[
    Matrix1 /. {Momentum[x1, D] → Momentum[X, D] - Momentum[x, D] / 2,
      Momentum[x2, D] → Momentum[X, D] + Momentum[x, D] / 2}]]
Coeff2 = Coeff1;
Phase2 = Expand[ExpandScalarProduct[
    Phase1 /. {Momentum[x1, D] → Momentum[X, D] - Momentum[x, D] / 2,
      Momentum[x2, D] → Momentum[X, D] + Momentum[x, D] / 2}]]
```

Out[28]=
$$\frac{i\,\mathrm{v2}\,\zeta\,(\gamma\cdot k).(\gamma\cdot a)\,a^2\,(k\cdot p)^2\,e^3}{m^7\,\chi\mathrm{p}^3}+\frac{i\,\mathrm{v1}\,\zeta\,(\gamma\cdot a).(\gamma\cdot k)\,a^2\,(k\cdot x)^2\,e^3}{8\,m^3\,\chi\mathrm{p}}-\frac{i\,\mathrm{v1}\,\zeta\,(\gamma\cdot a).(\gamma\cdot k)\,a^2\,(k\cdot X)^2\,e^3}{2\,m^3\,\chi\mathrm{p}}-$$
$$\frac{\mathrm{af}\,\zeta\,(\gamma\cdot a).(\gamma\cdot k)\,a^2\,(k\cdot p)\,(k\cdot x)\,e^3}{4\,m^5\,\chi\mathrm{p}^2}+\frac{\mathrm{af}\,\zeta\,(\gamma\cdot k).(\gamma\cdot a)\,a^2\,(k\cdot p)\,(k\cdot x)\,e^3}{4\,m^5\,\chi\mathrm{p}^2}+\frac{\mathrm{af}\,\zeta\,(\gamma\cdot a).(\gamma\cdot k)\,a^2\,(k\cdot p)\,(k\cdot X)\,e^3}{2\,m^5\,\chi\mathrm{p}^2}+$$
$$\frac{\mathrm{af}\,\zeta\,(\gamma\cdot k).(\gamma\cdot a)\,a^2\,(k\cdot p)\,(k\cdot X)\,e^3}{2\,m^5\,\chi\mathrm{p}^2}-\frac{\mathrm{af}\,\zeta\,a^2\,(k\cdot p)^2\,e^2}{m^5\,\chi\mathrm{p}^2}-\frac{\mathrm{v1}\,\gamma\cdot k\,a^2\,(k\cdot x)^2\,e^2}{8\,(k\cdot p)}+\frac{\mathrm{v1}\,\gamma\cdot k\,a^2\,(k\cdot X)^2\,e^2}{2\,(k\cdot p)}+$$
$$\frac{\mathrm{v2}\,\gamma\cdot k\,a^2\,(k\cdot p)\,e^2}{m^4\,\chi\mathrm{p}^2}+\frac{2\,\mathrm{tf}\,\zeta\,\gamma\cdot k\,a^2\,(k\cdot p)\,e^2}{m^4\,\chi\mathrm{p}^2}-\frac{\mathrm{af}\,\zeta\,(\gamma\cdot a).(\gamma\cdot k)\,(a\cdot p)\,(k\cdot p)\,e^2}{m^5\,\chi\mathrm{p}^2}-\frac{\mathrm{af}\,\zeta\,(\gamma\cdot k).(\gamma\cdot a)\,(a\cdot p)\,(k\cdot p)\,e^2}{m^5\,\chi\mathrm{p}^2}-$$
$$\frac{i\,\mathrm{sf}\,\zeta\,\gamma\cdot k\,a^2\,(k\cdot x)\,e^2}{2\,m^2\,\chi\mathrm{p}}-\frac{i\,\mathrm{af}\,\gamma\cdot k\,a^2\,(k\cdot x)\,e^2}{2\,m^2\,\chi\mathrm{p}}+\frac{i\,\mathrm{v1}\,\zeta\,(\gamma\cdot k).(\gamma\cdot p)\,a^2\,(k\cdot x)\,e^2}{4\,m^3\,\chi\mathrm{p}}+\frac{i\,\mathrm{v1}\,\zeta\,(\gamma\cdot p).(\gamma\cdot k)\,a^2\,(k\cdot x)\,e^2}{4\,m^3\,\chi\mathrm{p}}+$$
$$\frac{i\,\mathrm{v1}\,\zeta\,(\gamma\cdot a).(\gamma\cdot k)\,(a\cdot p)\,(k\cdot x)\,e^2}{2\,m^3\,\chi\mathrm{p}}+\frac{i\,\mathrm{v1}\,\zeta\,(\gamma\cdot k).(\gamma\cdot a)\,(a\cdot p)\,(k\cdot x)\,e^2}{2\,m^3\,\chi\mathrm{p}}+\frac{i\,\mathrm{v1}\,\zeta\,(\gamma\cdot k).(\gamma\cdot p)\,a^2\,(k\cdot X)\,e^2}{2\,m^3\,\chi\mathrm{p}}-$$
$$\frac{i\,\mathrm{v1}\,\zeta\,(\gamma\cdot p).(\gamma\cdot k)\,a^2\,(k\cdot X)\,e^2}{2\,m^3\,\chi\mathrm{p}}+\frac{i\,\mathrm{v1}\,\zeta\,(\gamma\cdot a).(\gamma\cdot k)\,(a\cdot p)\,(k\cdot X)\,e^2}{m^3\,\chi\mathrm{p}}+\frac{i\,\mathrm{v1}\,\zeta\,(\gamma\cdot k).(\gamma\cdot a)\,(a\cdot p)\,(k\cdot X)\,e^2}{m^3\,\chi\mathrm{p}}+$$
$$\frac{i\,\mathrm{tf}\,(\gamma\cdot a).(\gamma\cdot k)\,e}{m\,\chi\mathrm{p}}-\frac{i\,\mathrm{tf}\,(\gamma\cdot k).(\gamma\cdot a)\,e}{m\,\chi\mathrm{p}}-\frac{i\,\mathrm{sf}\,\zeta\,(\gamma\cdot a).(\gamma\cdot k).(\gamma\cdot p)\,e}{m^2\,\chi\mathrm{p}}-\frac{i\,\mathrm{af}\,(\gamma\cdot a).(\gamma\cdot k).(\gamma\cdot p)\,e}{m^2\,\chi\mathrm{p}}-\frac{i\,\mathrm{sf}\,\zeta\,\gamma\cdot k\,(a\cdot p)\,e}{m^2\,\chi\mathrm{p}}-$$
$$\frac{i\,\mathrm{af}\,\gamma\cdot k\,(a\cdot p)\,e}{m^2\,\chi\mathrm{p}}+\frac{2\,i\,\mathrm{v1}\,\zeta\,(\gamma\cdot k).(\gamma\cdot p)\,(a\cdot p)\,e}{m^3\,\chi\mathrm{p}}+\frac{i\,\mathrm{v1}\,\zeta\,(\gamma\cdot p).(\gamma\cdot k)\,(a\cdot p)\,e}{m^3\,\chi\mathrm{p}}+\frac{i\,\mathrm{sf}\,\zeta\,\gamma\cdot a\,(k\cdot p)\,e}{m^2\,\chi\mathrm{p}}+\frac{i\,\mathrm{af}\,\gamma\cdot a\,(k\cdot p)\,e}{m^2\,\chi\mathrm{p}}-$$
$$\frac{2\,i\,\mathrm{v1}\,\zeta\,(\gamma\cdot a).(\gamma\cdot p)\,(k\cdot p)\,e}{m^3\,\chi\mathrm{p}}-\frac{i\,\mathrm{v1}\,\zeta\,(\gamma\cdot p).(\gamma\cdot a)\,(k\cdot p)\,e}{m^3\,\chi\mathrm{p}}+\frac{m\,\mathrm{sf}\,(\gamma\cdot a).(\gamma\cdot k)\,(k\cdot x)\,e}{4\,(k\cdot p)}-\frac{m\,\mathrm{sf}\,(\gamma\cdot k).(\gamma\cdot a)\,(k\cdot x)\,e}{4\,(k\cdot p)}+$$
$$\frac{\mathrm{v1}\,(\gamma\cdot k).(\gamma\cdot a).(\gamma\cdot p)\,(k\cdot x)\,e}{4\,(k\cdot p)}-\frac{\mathrm{v1}\,(\gamma\cdot p).(\gamma\cdot a).(\gamma\cdot k)\,(k\cdot x)\,e}{4\,(k\cdot p)}-\frac{m\,\mathrm{sf}\,(\gamma\cdot a).(\gamma\cdot k)\,(k\cdot X)\,e}{2\,(k\cdot p)}-\frac{m\,\mathrm{sf}\,(\gamma\cdot k).(\gamma\cdot a)\,(k\cdot X)\,e}{2\,(k\cdot p)}+$$
$$\frac{\mathrm{v1}\,(\gamma\cdot k).(\gamma\cdot a).(\gamma\cdot p)\,(k\cdot X)\,e}{2\,(k\cdot p)}+\frac{\mathrm{v1}\,(\gamma\cdot p).(\gamma\cdot a).(\gamma\cdot k)\,(k\cdot X)\,e}{2\,(k\cdot p)}+\frac{i\,\mathrm{v1}\,\zeta\,(\gamma\cdot a).(\gamma\cdot k)\,p^2\,e}{m^3\,\chi\mathrm{p}}+m\,\mathrm{sf}-\mathrm{v1}\,\gamma\cdot p$$

Out[30]=
$$\frac{a^2\,e^2\,(k\cdot x)\,(k\cdot X)^2}{2\,(k\cdot p)}+\frac{a^2\,e^2\,(k\cdot x)^3}{24\,(k\cdot p)}+\frac{e\,(a\cdot p)\,(k\cdot x)\,(k\cdot X)}{k\cdot p}-p\cdot x$$

In[31]:=
```
Matrix3 =
  (((Expand[Matrix2 /. TripleGamma]) /. EpsToF) /. FieldSubstitutions) /. akToF
Coeff3 = Coeff2;
Phase3 = Phase2;
```

Out[31]=

$$\frac{\text{v2}\,\zeta\,\sigma\text{F}\,a^2\,(k\cdot p)^2\,e^3}{2\,m^7\,\chi\text{p}^3}-\frac{\text{v1}\,\zeta\,\sigma\text{F}\,a^2\,(k\cdot x)^2\,e^3}{16\,m^3\,\chi\text{p}}+\frac{\text{v1}\,\zeta\,\sigma\text{F}\,a^2\,(k\cdot X)^2\,e^3}{4\,m^3\,\chi\text{p}}-\frac{i\,\text{af}\,\zeta\,\sigma\text{F}\,a^2\,(k\cdot p)\,(k\cdot x)\,e^3}{4\,m^5\,\chi\text{p}^2}-$$
$$\frac{\text{af}\,\zeta\,a^2\,(k\cdot p)^2\,e^2}{m^5\,\chi\text{p}^2}-\frac{\text{v1}\,\gamma\cdot k\,a^2\,(k\cdot x)^2\,e^2}{8\,(k\cdot p)}+\frac{\text{v1}\,\gamma\cdot k\,a^2\,(k\cdot X)^2\,e^2}{2\,(k\cdot p)}+\frac{\text{v2}\,\gamma\cdot k\,a^2\,(k\cdot p)\,e^2}{m^4\,\chi\text{p}^2}+$$
$$\frac{2\,\text{tf}\,\zeta\,\gamma\cdot k\,a^2\,(k\cdot p)\,e^2}{m^4\,\chi\text{p}^2}-\frac{i\,\text{sf}\,\zeta\,\gamma\cdot k\,a^2\,(k\cdot x)\,e^2}{2\,m^2\,\chi\text{p}}-\frac{i\,\text{af}\,\gamma\cdot k\,a^2\,(k\cdot x)\,e^2}{2\,m^2\,\chi\text{p}}+\frac{i\,\text{v1}\,\zeta\,(\gamma\cdot k).(\gamma\cdot p)\,a^2\,(k\cdot x)\,e^2}{4\,m^3\,\chi\text{p}}+$$
$$\frac{i\,\text{v1}\,\zeta\,(\gamma\cdot p).(\gamma\cdot k)\,a^2\,(k\cdot x)\,e^2}{4\,m^3\,\chi\text{p}}+\frac{i\,\text{v1}\,\zeta\,(\gamma\cdot k).(\gamma\cdot p)\,a^2\,(k\cdot X)\,e^2}{2\,m^3\,\chi\text{p}}-\frac{i\,\text{v1}\,\zeta\,(\gamma\cdot p).(\gamma\cdot k)\,a^2\,(k\cdot X)\,e^2}{2\,m^3\,\chi\text{p}}+$$
$$\frac{\text{sf}\,\zeta\,\gamma^\beta.\overline{\gamma}^5\,\text{FDp}^\beta\,e}{m^2\,\chi\text{p}}+\frac{\text{af}\,\gamma^\beta.\overline{\gamma}^5\,\text{FDp}^\beta\,e}{m^2\,\chi\text{p}}+\frac{2\,i\,\text{v1}\,\zeta\,(\gamma\cdot k).(\gamma\cdot p)\,(a\cdot p)\,e}{m^3\,\chi\text{p}}+\frac{i\,\text{v1}\,\zeta\,(\gamma\cdot p).(\gamma\cdot k)\,(a\cdot p)\,e}{m^3\,\chi\text{p}}-$$
$$\frac{2\,i\,\text{v1}\,\zeta\,(\gamma\cdot a).(\gamma\cdot p)\,(k\cdot p)\,e}{m^3\,\chi\text{p}}-\frac{i\,\text{v1}\,\zeta\,(\gamma\cdot p).(\gamma\cdot a)\,(k\cdot p)\,e}{m^3\,\chi\text{p}}+\frac{i\,m\,\text{sf}\,\sigma\text{F}\,(k\cdot x)\,e}{4\,(k\cdot p)}-\frac{i\,\text{v1}\,\gamma^\beta.\overline{\gamma}^5\,\text{FDp}^\beta\,(k\cdot x)\,e}{2\,(k\cdot p)}-$$
$$\text{v1}\,\gamma\cdot a\,(k\cdot X)\,e+\frac{\text{v1}\,\gamma\cdot k\,(a\cdot p)\,(k\cdot X)\,e}{k\cdot p}-\frac{\text{v1}\,\zeta\,\sigma\text{F}\,p^2\,e}{2\,m^3\,\chi\text{p}}-\frac{\text{tf}\,\sigma\text{F}\,e}{m\,\chi\text{p}}+m\,\text{sf}-\text{v1}\,\gamma\cdot p$$

In[34]:=
```
Matrix4 = Contract[
  DiracOrder[
    Matrix3 /. {DiracGamma[LorentzIndex[β, D], D].DiracGamma[5] ×
        Pair[LorentzIndex[β, D], Momentum[FDp, D]] → FVD[γγ5FD, α] × pv[α]}
   ] /. {
    DiracGamma[Momentum[a, D], D].
      DiracGamma[Momentum[p, D], D] → FVD[γaγ, α] × pv[α],
    DiracGamma[Momentum[k, D], D].
      DiracGamma[Momentum[p, D], D] → FVD[γkγ, α] × pv[α]}
 ]
Coeff4 = Coeff3;
Phase4 = Phase3;
```

Out[34]=

$$-\frac{1}{16\, m^7\, \chi\text{p}^3\, (k\cdot p)}\Big(4\, i\, a^2\, \text{af}\, e^3\, \zeta\, m^2\, \sigma\text{F}\, \chi\text{p}\, (k\cdot x)\, (k\cdot p)^2 + 16\, a^2\, \text{af}\, e^2\, \zeta\, m^2\, \chi\text{p}\, (k\cdot p)^3 + a^2\, e^3\, \zeta\, m^4\, \sigma\text{F}\, \text{v1}\, \chi\text{p}^2\, (k\cdot p)\, (k\cdot x)^2 - 4\, a^2\, e^3\, \zeta\, m^4\, \sigma\text{F}\, \text{v1}\, \chi\text{p}^2\, (k\cdot p)\, (k\cdot X)^2 - 8\, a^2\, e^3\, \zeta\, \sigma\text{F}\, \text{v2}\, (k\cdot p)^3 - 4\, i\, e\, m^8\, \text{sf}\, \sigma\text{F}\, \chi\text{p}^3\, (k\cdot x) + 16\, e\, m^6\, \sigma\text{F}\, \text{tf}\, \chi\text{p}^2\, (k\cdot p) + 8\, e\, \zeta\, m^4\, p^2\, \sigma\text{F}\, \text{v1}\, \chi\text{p}^2\, (k\cdot p) - 16\, m^8\, \text{sf}\, \chi\text{p}^3\, (k\cdot p)\Big) + \frac{1}{8\, m^4\, \chi\text{p}^2\, (k\cdot p)}\, e\, \gamma\cdot k\, \Big(a^2\, (-e)\, m^4\, \text{v1}\, \chi\text{p}^2\, (k\cdot x)^2 + 4\, a^2\, e\, m^4\, \text{v1}\, \chi\text{p}^2\, (k\cdot X)^2 + 16\, a^2\, e\, \zeta\, \text{tf}\, (k\cdot p)^2 + 8\, a^2\, e\, \text{v2}\, (k\cdot p)^2 - 4\, i\, a^2\, \text{af}\, e\, m^2\, \chi\text{p}\, (k\cdot p)\, (k\cdot x) - 4\, i\, a^2\, e\, \zeta\, m^2\, \text{sf}\, \chi\text{p}\, (k\cdot p)\, (k\cdot x) + 8\, m^4\, \text{v1}\, \chi\text{p}^2\, (a\cdot p)\, (k\cdot X)\Big) - \frac{i\, e\, \zeta\, \text{v1}\, (k\cdot p)\, (2\, (a\cdot p) - p\cdot \gamma\text{a}\gamma)}{m^3\, \chi\text{p}} + \frac{i\, e\, \zeta\, \text{v1}\, (2\, (k\cdot p) - p\cdot \gamma\text{k}\gamma)\, \big(a^2\, e\, (k\cdot x) - 2\, a^2\, e\, (k\cdot X) + 4\, (a\cdot p)\big)}{4\, m^3\, \chi\text{p}} + \frac{i\, e\, \zeta\, \text{v1}\, (p\cdot \gamma\text{k}\gamma)\, \big(a^2\, e\, (k\cdot x) + 2\, a^2\, e\, (k\cdot X) + 8\, (a\cdot p)\big)}{4\, m^3\, \chi\text{p}} - e\, \text{v1}\, \gamma\cdot a\, (k\cdot X) + \frac{e\, (p\cdot \gamma\gamma\text{5FD})\, \big(2\, \text{af}\, (k\cdot p) - i\, m^2\, \text{v1}\, \chi\text{p}\, (k\cdot x) + 2\, \zeta\, \text{sf}\, (k\cdot p)\big)}{2\, m^2\, \chi\text{p}\, (k\cdot p)} - \frac{2\, i\, e\, \zeta\, \text{v1}\, (k\cdot p)\, (p\cdot \gamma\text{a}\gamma)}{m^3\, \chi\text{p}} - \text{v1}\, \gamma\cdot p$$

Expanding scalar products into components and changing variables

$p \to \{p_- = 1/2\,(p^0 - p^3),\ p_+ = p^0 + p^3,\ p_\perp\}$

$p_- = x_-/2\,s$

$p_+ = (p^2 + p_\perp{}^2)/2\,p_- = s\,(p^2 + p_\perp{}^2)/x_-$

Integration measure

$\int \mathbb{d}^D \mathbf{p} \ldots = \int \frac{\mathbb{d}s}{2\,s}\,\mathbb{d}p^2\,\mathbb{d}^{D-2}p_\perp$

$x_- = kx/m$

$p_- = kx/2\,m\,s$

$kp = m\,p_- = m\,xm/2\,s = kx/2\,s$

$ap = -at\,pt$

$\gamma p = \gamma_-\,p_+ + \gamma_+\,p_- - \gamma_\perp\,p_\perp = Gm\,\frac{s}{x_-}\,(p^2 + p_\perp{}^2) + Gp\,\frac{x_-}{2\,s} - Gt\,pt$

$\gamma k = \gamma_-\,k_+ = m\,Gm$

$kx = k_+\,x_- = m\,xm$

$(\gamma F^*)^\mu.\gamma^5 \to \{(\gamma F^*)_-.\gamma^5 = 0,\ (\gamma F^*)_+.\gamma^5,\ (\gamma F^*)_\perp.\gamma^5\} = \{0,\ \gamma\gamma 5FDp,\ \gamma\gamma 5FDt\}$

$(\gamma F^*)_\mu\,k^\mu = 0 \to (\gamma F^*)_- = 0$

$(\gamma F^*)_\mu\,a^\mu = 0 \to \gamma\gamma 5FDt\,at = 0$

$(\gamma F^*)^\mu.\gamma^5\,p_\mu = \gamma\gamma 5FDm * pp$

$\gamma k\gamma_- = (\gamma k)^2/m = 0$

In[37]:=
```
D[{xm / 2 / s, s (p2 + pt2) / xm}, {{s, p2}}]
Abs[Det[%]]
```

Out[37]= $\begin{pmatrix} -\frac{xm}{2\,s^2} & 0 \\ \frac{p2+pt2}{xm} & \frac{s}{xm} \end{pmatrix}$

Out[38]= $\frac{1}{2\,|s|}$

In[39]:=
```
Matrix5 = Collect[
  Expand[Matrix4 /. {DiracGamma[Momentum[p, D], D] → Gp * pm + Gm * pp - Gt[i] * pt[i],
        Pair[Momentum[a, D], Momentum[p, D]] → -at[i1] × pt[i1],
        Pair[Momentum[p, D], Momentum[γγ5FD, D]] → γγ5FDp * pm,
        Pair[Momentum[p, D], Momentum[γaγ, D]] →
         γaγp * pm + γaγm * pp - γaγt[i] * pt[i],
        Pair[Momentum[p, D], Momentum[γkγ, D]] → γkγp * pm - γkγt[i] * pt[i],
        (*av2→-at^2,*)
        DiracGamma[Momentum[k, D], D] → m Gm} /. {γkγm → 0} /.
      {χp → ξ kp / m^2} /. {kp → kx / 2 / s, pm → xm / 2 / s} /. {kx → m xm} /.
    {pp → s ( pv2 + pt[i2] ^ 2) / xm}], {v1, pt[i2] × pt[i], pt[i2], pt[i_],
   pv2, γγ5FDm, γγ5FDp, γγ5FDt[i], γkγm, γkγp, γkγt[i], γaγm, γaγp, γaγt[i]}]
Coefficient[Matrix5, γkγt[i]]
Coeff5 = Coeff4 / 2 / s
Phase5 = Collect[
  Expand[Phase4 /. {Pair[Momentum[p, D], Momentum[x, D]] → pp xm + pm xp - pt * xt,
      kp → m pm, Pair[Momentum[a, D], Momentum[p, D]] → -at pt } /.
     {kp → kx / 2 / s, pm → xm / 2 / s} /. {pp → s ( pv2 + pt^2) / xm} /. {xm →
     kx / m}], {pt, pp}]
```

Out[39]=
$$\mathrm{v1}\left(\frac{a^2 e^3 \zeta\, s\, \sigma\mathrm{F}\,(k\cdot X)^2}{2 m^2 \xi\, \mathrm{xm}}+\frac{a^2 e^2\, \mathrm{Gm}\, s\,(k\cdot X)^2}{\mathrm{xm}}+\right.$$
$$\mathrm{pt}(i)\left(-\frac{2 i a^2 e^2 \zeta\, s\, \gamma\mathrm{k}\gamma\mathrm{t}(i)\,(k\cdot X)}{m^2 \xi\, \mathrm{xm}}+\frac{2 i e \zeta\, s\, \mathrm{at(i1)}\,\gamma\mathrm{k}\gamma\mathrm{t}(i)\,\mathrm{pt(i1)}}{m^2 \xi\, \mathrm{xm}}+\frac{i e \zeta\, \gamma\mathrm{a}\gamma\mathrm{t}(i)}{m \xi}+\mathrm{Gt}(i)\right)-$$
$$\frac{a^2 e^3 \zeta\, s\, \sigma\mathrm{F}\, \mathrm{xm}}{8 \xi}-\frac{1}{4} a^2 e^2\, \mathrm{Gm}\, m^2 s\, \mathrm{xm}+\frac{i a^2 \gamma\mathrm{k}\gamma\mathrm{p}\, e^2 \zeta\,(k\cdot X)}{m^2 \xi}-\frac{i a^2 e^2 \zeta\,(k\cdot X)}{m \xi}+\frac{i a^2 e^2 \zeta\, \mathrm{xm}}{2 \xi}-$$
$$e\, \gamma\cdot a\,(k\cdot X)+\mathrm{pt(i1)}\left(-\frac{2 e\, \mathrm{Gm}\, s\, \mathrm{at(i1)}\,(k\cdot X)}{\mathrm{xm}}-\frac{i \gamma\mathrm{k}\gamma\mathrm{p}\, e \zeta\, \mathrm{at(i1)}}{m^2 \xi}\right)+\mathrm{pt(i2)}^2\left(-\frac{\mathrm{Gm}\, s}{\mathrm{xm}}-\frac{i \gamma\mathrm{a}\gamma\mathrm{m}\, e \zeta\, s}{m \xi\, \mathrm{xm}}\right)+$$
$$\left.p^2\left(-\frac{e \zeta\, s\, \sigma\mathrm{F}}{m^2 \xi\, \mathrm{xm}}-\frac{i \gamma\mathrm{a}\gamma\mathrm{m}\, e \zeta\, s}{m \xi\, \mathrm{xm}}-\frac{\mathrm{Gm}\, s}{\mathrm{xm}}\right)-\frac{i \gamma\mathrm{a}\gamma\mathrm{p}\, e \zeta\, \mathrm{xm}}{2 m \xi\, s}-\frac{1}{2} i \gamma\gamma5\mathrm{FDp}\, e\, \mathrm{xm}-\frac{\mathrm{Gp}\, \mathrm{xm}}{2 s}\right)-$$
$$\frac{i a^2\, \mathrm{af}\, e^3 \zeta\, s\, \sigma\mathrm{F}}{2 m \xi^2}-\frac{i a^2\, \mathrm{af}\, e^2\, \mathrm{Gm}\, m\, s}{\xi}-\frac{a^2\, \mathrm{af}\, e^2 \zeta}{m \xi^2}+\frac{a^2 e^3 \zeta\, s\, \sigma\mathrm{F}\, \mathrm{v2}}{m^2 \xi^3\, \mathrm{xm}}-$$
$$\frac{i a^2 e^2 \zeta\, \mathrm{Gm}\, m\, s\, \mathrm{sf}}{\xi}+$$
$$\frac{4 a^2 e^2 \zeta\, \mathrm{Gm}\, s\, \mathrm{tf}}{\xi^2\, \mathrm{xm}}+$$
$$\frac{2 a^2 e^2\, \mathrm{Gm}\, s\, \mathrm{v2}}{\xi^2\, \mathrm{xm}}+\gamma\gamma5\mathrm{FDp}\left(\frac{\mathrm{af}\, e}{m \xi}+\frac{e \zeta\, \mathrm{sf}}{m \xi}\right)+$$
$$\frac{1}{2} i e\, m\, s\, \mathrm{sf}\, \sigma\mathrm{F}-\frac{2 e\, s\, \sigma\mathrm{F}\, \mathrm{tf}}{\xi\, \mathrm{xm}}+m\, \mathrm{sf}$$

Out[40]=
$$\frac{2 i e \zeta\, s\, \mathrm{v1}\, \mathrm{at(i1)}\, \mathrm{pt}(i)\, \mathrm{pt(i1)}}{m^2 \xi\, \mathrm{xm}}-\frac{2 i a^2 e^2 \zeta\, s\, \mathrm{v1}\, \mathrm{pt}(i)\,(k\cdot X)}{m^2 \xi\, \mathrm{xm}}$$

Out[41]= $$\frac{i\,2^{-D-2}\,\pi^{-D}\,\Lambda^{4-D}}{s\,\mathrm{DdInv}(\zeta)}$$

Out[42]= $$\frac{1}{12}\,a^2\,e^2\,s\,(\,k\cdot x\,)^2+a^2\,e^2\,s\,(\,k\cdot X\,)^2+\mathrm{pt}\,(\mathrm{xt}-2\,\mathrm{at}\,e\,s\,(k\cdot X))-\frac{\mathrm{xp}\,(k\cdot x)}{2\,m\,s}-p^2\,s+\mathrm{pt}^2\,(-s)$$

Integration over

$\int \mathbb{d}^{D-2} p_{\perp} \ldots$

$$I_0 = \int d^{D-2} p_{\perp} \, \mathrm{Exp}\left[-I\, A\, p_{\perp}^2 + I\, (J.p_{\perp})\right] =$$

$$= \mathrm{Exp}\left[-I \frac{\pi}{2} \frac{D-2}{2}\right] \pi^{\frac{D-2}{2}} (\det A)^{-\frac{1}{2}} \, \mathrm{Exp}\left[I \frac{1}{4} J.A^{-1}.J\right]$$

$$I_{1\,i} = \int d^{D-2} p_{\perp} \, p_{\perp i} \, \mathrm{Exp}\left[-I\, A\, p_{\perp}^2 + I\, (J.p_{\perp})\right] =$$

$$= \frac{1}{2} \left(A^{-1}.J\right)_i I_0$$

$$I_2 = \int d^{D-2} p_{\perp} \, p_{\perp}{}^2 \, \mathrm{Exp}\left[-I\, A\, p_{\perp}^2 + I\, (J.p_{\perp})\right] =$$

$$= \left[-\mathbb{i} \frac{1}{2} \mathrm{Tr}\, A^{-1} + \left(\frac{1}{2} A^{-1}.J\right)^2\right]$$

$$I_{2\,ij} = \int d^{D-2} p_{\perp} \, p_{\perp i} \, p_{\perp j} \, \mathrm{Exp}\left[-I\, A\, p_{\perp}^2 + I (J.p_{\perp})\right] =$$

$$= \left[-\mathbb{i} \frac{1}{2} A^{-1}{}_{ij} + \left(\frac{1}{2} A^{-1}.J\right)_i \left(\frac{1}{2} A^{-1}.J\right)_j\right] I_0$$

$$I_{3\,i} = -\mathbb{i} \frac{\partial}{\partial J_i} I_2 = \int d^{D-2} p_{\perp} \, p_{\perp}{}^2 \, p_{\perp i} \, \mathrm{Exp}\left[-I\, A\, p_{\perp}^2 + I (J.p_{\perp})\right] =$$

$$= \left\{\left[-\mathbb{i} \frac{1}{2} \mathrm{Tr}\, A^{-1} + \left(\frac{1}{2} A^{-1}.J\right)^2\right] \frac{1}{2} \left(A^{-1}.J\right)_i - \mathbb{i} \frac{1}{2} \left(A^{-1\,T} A^{-1}.J\right)_i \right\} I_0 =$$

$$= \left(\frac{1}{2} A^{-1}.J\right)_i I_2 - \mathbb{i} \left(A^{-1\,T}\right)_{ij} I_{1\,j}$$

where

$A = s,$

$J = x_{\perp} - 2\, e a_{\perp}\, s\, kX,$

$\det A = s^{D-2},$

$A^{-1} = 1 / s$

We perform integrations

Integrations changes the coefficient (Coeff) and phase

Then recollect some scalar products

$a_{\perp}^2 = -a^2$

$a_{\perp}\, x_{\perp} = -(ax)$

$x_{\perp}^2 = 2\, x_{-}\, x_{+} - x^2$

In[43]:=
```
Clear[J]
Amatr = -Coefficient[Phase5, pt^2]
J[i_] = Coefficient[Phase5, pt] /. {at → at[i], xt → xt[i]}
CI0 = Exp[-I Pi / 2 (D / 2 - 1)] Pi^(D / 2 - 1) / Amatr^(D / 2 - 1)
```

Out[44]= s

Out[45]= $\mathrm{xt}(i) - 2\, e\, s\, \mathrm{at}(i)\, (k \cdot X)$

Out[46]= $e^{-\frac{1}{2} i \pi \left(\frac{D}{2}-1\right)} \pi^{\frac{D}{2}-1} s^{1-\frac{D}{2}}$

In[47]:=
```
Phase6 = Expand[Expand[((Phase5 /. {pt → 0}) + 1 / 4 J[i]^2 / Amatr)] /.
    { at[i]^2 → -av2, at[i] × xt[i] → -ax, xt[i]^2 → 2 xm xp - xv2} /. {xm → kx / m}]
Coeff6 = Coeff5 * CI0
Expand[
  Expand[Matrix5] /.
      {pt[i2_]^2 × pt[i1_] → ((-I / 2 * (D - 2) / Amatr + ( 1 / 2 / Amatr * J[i2])^2) *
            1 / 2 / Amatr * J[i1] - I * 1 / 2 / Amatr^2 * J[i1])} /. {pt[i_] × pt[i1_] →
        (- I / 2 / Amatr δ[i, i1] + (1 / 2 / Amatr * J[i]) * (1 / 2 / Amatr * J[i1]))} /.
    {pt[i_]^2 → (-I / 2 * (D - 2) / Amatr + ( 1 / 2 / Amatr * J[i])^2) } /.
   {pt[i_] → ( 1 / 2 / Amatr * J[i])}
 ];
% - Coefficient[%, δ[i, i1]] δ[i, i1] + (Coefficient[%, δ[i, i1]] /. {i1 → i});
Matrix6 = Collect[
  % /. {γγ5FDt[i_] at[i_] → 0 } /. { at[i_]^2 → -av2} /. {at[i_] × xt[i_] → -ax} /.
    { xt[i_]^2 → 2 xm xp - xv2} /. {at[i_] ax kX xt[i_] → -ax ax kX},
  {v1, pt[i2] × pt[i], pt[i2], pt[i_], pv2, γγ5FDm,
   γγ5FDp, γγ5FDt[i], γkγm, γkγp, γkγt[i], γaγm, γaγp, γaγt[i]}]
(*Matrix6=Collect[Expand[Expand[(Matrix5/.{pt[i_]→0})+
      Coefficient[Matrix5,pt]*1/2/Amatr*J+
      Coefficient[Matrix5,pt^2]*(-I/2*(D-2)/Amatr+( 1/2/Amatr*J)^2)+
      Coefficient[Matrix5,pt^3]
       ((-I/2*(D-2)/Amatr+( 1/2/Amatr*J)^2)*1/2/Amatr*J+1/2/Amatr^2*J)]/.
    {γγ5FDt at→ 0, at^4→ av2^2 ,at^3→ -av2 at, at^2→-av2,
     at xt→ -ax, xt^2→ 2xm xp -xv2}],
  {p2,Gm,Gp,Gt,γγ5FDm,γγ5FDp,γγ5FDt,γkγm,γkγp,γkγt,γaγm,γaγp,γaγt}]*)
```

Out[47]= $\frac{1}{12} a^2 e^2 s\, (k \cdot x)^2 + e\, (a \cdot x)\, (k \cdot X) - p^2 s - \frac{x^2}{4 s}$

Out[48]= $\frac{i\, 2^{-D-2} e^{-\frac{1}{2} i \pi \left(\frac{D}{2}-1\right)} \pi^{-\frac{D}{2}-1} \Lambda^{4-D} s^{-D/2}}{\mathrm{DdInv}(\zeta)}$

Out[51]=

$$\text{v1}\left(\gamma\text{a}\gamma\text{m}\left(\frac{i\,a^2\,e^3\,\zeta\,s\,(k\cdot X)^2}{m\,\xi\,\text{xm}}-\frac{i\,e^2\,\zeta\,(a\cdot x)\,(k\cdot X)}{m\,\xi\,\text{xm}}-\frac{D\,e\,\zeta}{2\,m\,\xi\,\text{xm}}-\frac{i\,e\,\zeta\left(2\,\text{xm}\,\text{xp}-x^2\right)}{4\,m\,\xi\,s\,\text{xm}}+\frac{e\,\zeta}{m\,\xi\,\text{xm}}\right)+\right.$$

$$\frac{a^2\,e^3\,\zeta\,s\,\sigma\text{F}\,(k\cdot X)^2}{2\,m^2\,\xi\,\text{xm}}+\gamma\text{k}\gamma\text{t}(i)\left(\frac{i\,e^2\,\zeta\,\text{at}(i)\,(a\cdot x)\,(k\cdot X)}{m^2\,\xi\,\text{xm}}-\frac{i\,e\,\zeta\,\text{xt}(i)\,(a\cdot x)}{2\,m^2\,\xi\,s\,\text{xm}}+\frac{e\,\zeta\,\text{at}(i)}{m^2\,\xi\,\text{xm}}\right)-\frac{a^2\,e^3\,\zeta\,s\,\sigma\text{F}\,\text{xm}}{8\,\xi}-$$

$$\frac{1}{4}\,a^2\,e^2\,\text{Gm}\,m^2\,s\,\text{xm}-\frac{i\,a^2\,e^2\,\zeta\,(k\cdot X)}{m\,\xi}+\frac{i\,a^2\,e^2\,\zeta\,\text{xm}}{2\,\xi}-e\,\gamma\cdot a\,(k\cdot X)+\frac{i\,\gamma\text{k}\gamma\text{p}\,e\,\zeta\,(a\cdot x)}{2\,m^2\,\xi\,s}+$$

$$\gamma\text{a}\gamma\text{t}(i)\left(\frac{i\,e\,\zeta\,\text{xt}(i)}{2\,m\,\xi\,s}-\frac{i\,e^2\,\zeta\,\text{at}(i)\,(k\cdot X)}{m\,\xi}\right)-e\,\text{at}(i)\,\text{Gt}(i)\,(k\cdot X)+\frac{i\,D\,\text{Gm}}{2\,\text{xm}}+p^2\left(-\frac{e\,\zeta\,s\,\sigma\text{F}}{m^2\,\xi\,\text{xm}}-\frac{i\,\gamma\text{a}\gamma\text{m}\,e\,\zeta\,s}{m\,\xi\,\text{xm}}-\frac{\text{Gm}\,s}{\text{xm}}\right)-$$

$$\left.\frac{i\,\gamma\text{a}\gamma\text{p}\,e\,\zeta\,\text{xm}}{2\,m\,\xi\,s}-\frac{1}{2}\,i\,\gamma\gamma5\text{FDp}\,e\,\text{xm}-\frac{\text{Gm}\left(2\,\text{xm}\,\text{xp}-x^2\right)}{4\,s\,\text{xm}}-\frac{i\,\text{Gm}}{\text{xm}}-\frac{\text{Gp}\,\text{xm}}{2\,s}+\frac{\text{Gt}(i)\,\text{xt}(i)}{2\,s}\right)-$$

$$\frac{i\,a^2\,\text{af}\,e^3\,\zeta\,s\,\sigma\text{F}}{2\,m\,\xi^2}-\frac{i\,a^2\,\text{af}\,e^2\,\text{Gm}\,m\,s}{\xi}-\frac{a^2\,\text{af}\,e^2\,\zeta}{m\,\xi^2}+\frac{a^2\,e^3\,\zeta\,s\,\sigma\text{F}\,\text{v2}}{m^2\,\xi^3\,\text{xm}}-$$

$$\frac{i\,a^2\,e^2\,\zeta\,\text{Gm}\,m\,s\,\text{sf}}{\xi}+$$

$$\frac{4\,a^2\,e^2\,\zeta\,\text{Gm}\,s\,\text{tf}}{\xi^2\,\text{xm}}+\frac{2\,a^2\,e^2\,\text{Gm}\,s\,\text{v2}}{\xi^2\,\text{xm}}+$$

$$\gamma\gamma5\text{FDp}\left(\frac{\text{af}\,e}{m\,\xi}+\frac{e\,\zeta\,\text{sf}}{m\,\xi}\right)+$$

$$\frac{1}{2}\,i\,e\,m\,s\,\text{sf}\,\sigma\text{F}-\frac{2\,e\,s\,\sigma\text{F}\,\text{tf}}{\xi\,\text{xm}}+m\,\text{sf}$$

Next we substitute

$\gamma_\perp\, a_\perp = -\gamma a$

$\gamma_\perp\, x_\perp = \gamma_-\, x_+ + \gamma_+\, x_- - \gamma x$

$\gamma_-\, x_- = \frac{\gamma k}{m}\, x_-$

$x_- = kx\,/\,m$

In[52]:=
```
Matrix7 = Collect[
  DiracSimplify[
   Expand[Matrix6] /. {Gt[i_] × at[i_] → -Contract[GAD[α] × av[α]],
         Gt[i_] × xt[i_] → Gm xp + Gp xm - Contract[GAD[α] × xv[α]] ,
         (*γγ5FDt[i_] xt[i_] →
          γγ5FDp xm - DiracSlash[FDx,Dimension→D].GA[5],*)
         γγ5FDp  → DiracSlash[FDx, Dimension → D].GA[5] / xm,
         γkγt[i_] xt[i_] → -Pair[Momentum[x, D], Momentum[γkγ, D]] + γkγp * xm,
         γkγt[i_] at[i_] → -Pair[Momentum[a, D], Momentum[γkγ, D]],
         γaγt[i_] xt[i_] →
          - Pair[Momentum[x, D], Momentum[γaγ, D]] + γaγp * xm + γaγm * xp,
         γaγt[i_] at[i_]  → - Pair[Momentum[a, D], Momentum[γaγ, D]]} /.
      {Gm → DiracGamma[Momentum[k, D], D] / m } /. {γkγm → 0} /.
    {γaγm → DiracGamma[Momentum[a, D], D].DiracGamma[Momentum[k, D], D] / m }
   /. {xm → kx / m}
  ],
  {v1, p2, Gm, Gp, Gt, γγ5FDm, γγ5FDp, γγ5FDt, γkγm, γkγp, γkγt, γaγm, γaγp, γaγt}]
Coeff7 = Coeff6
Phase7 = Phase6
```

Out[52]=

$$\begin{aligned}
&-\frac{i\,\mathrm{af}\,s\,\zeta\,\sigma\mathrm{F}\,a^2\,e^3}{2\,m\,\xi^2}+\frac{s\,\mathrm{v2}\,\zeta\,\sigma\mathrm{F}\,a^2\,e^3}{m\,\xi^3\,(k\cdot x)}-\frac{i\,\mathrm{af}\,s\,\gamma\cdot k\,a^2\,e^2}{\xi}-\frac{i\,s\,\mathrm{sf}\,\zeta\,\gamma\cdot k\,a^2\,e^2}{\xi}-\frac{\mathrm{af}\,\zeta\,a^2\,e^2}{m\,\xi^2}+\\
&\frac{2\,s\,\mathrm{v2}\,\gamma\cdot k\,a^2\,e^2}{\xi^2\,(k\cdot x)}+\frac{4\,s\,\mathrm{tf}\,\zeta\,\gamma\cdot k\,a^2\,e^2}{\xi^2\,(k\cdot x)}+\frac{1}{2}\,i\,m\,s\,\mathrm{sf}\,\sigma\mathrm{F}\,e-\frac{2\,m\,s\,\mathrm{tf}\,\sigma\mathrm{F}\,e}{\xi\,(k\cdot x)}+\frac{\mathrm{af}\,(\gamma\cdot\mathrm{FDx}).\bar{\gamma}^5\,e}{\xi\,(k\cdot x)}+\\
&\frac{\mathrm{sf}\,\zeta\,(\gamma\cdot\mathrm{FDx}).\bar{\gamma}^5\,e}{\xi\,(k\cdot x)}+m\,\mathrm{sf}+\mathrm{v1}\left(\frac{s\,\zeta\,\sigma\mathrm{F}\,a^2\,(k\cdot X)^2\,e^3}{2\,m\,\xi\,(k\cdot x)}+\frac{i\,s\,\zeta\,(\gamma\cdot a).(\gamma\cdot k)\,a^2\,(k\cdot X)^2\,e^3}{m\,\xi\,(k\cdot x)}-\right.\\
&\quad\frac{s\,\zeta\,\sigma\mathrm{F}\,a^2\,(k\cdot x)\,e^3}{8\,m\,\xi}-\frac{1}{4}\,s\,\gamma\cdot k\,a^2\,(k\cdot x)\,e^2+\frac{i\,\zeta\,a^2\,(k\cdot x)\,e^2}{2\,m\,\xi}-\frac{i\,\zeta\,a^2\,(k\cdot X)\,e^2}{m\,\xi}+\frac{i\,\zeta\,(a\cdot\gamma\mathrm{a}\gamma)\,(k\cdot X)\,e^2}{m\,\xi}-\\
&\quad\frac{i\,\zeta\,(\gamma\cdot a).(\gamma\cdot k)\,(a\cdot x)\,(k\cdot X)\,e^2}{m\,\xi\,(k\cdot x)}-\frac{i\,\zeta\,(a\cdot x)\,(a\cdot\gamma\mathrm{k}\gamma)\,(k\cdot X)\,e^2}{m\,\xi\,(k\cdot x)}-\frac{1}{2}\,i\,(\gamma\cdot\mathrm{FDx}).\bar{\gamma}^5\,e-\frac{s\,\zeta\,\sigma\mathrm{F}\,p^2\,e}{m\,\xi\,(k\cdot x)}-\\
&\quad\frac{i\,s\,\zeta\,(\gamma\cdot a).(\gamma\cdot k)\,p^2\,e}{m\,\xi\,(k\cdot x)}+\frac{i\,\zeta\,(\gamma\cdot a).(\gamma\cdot k)\,x^2\,e}{4\,m\,s\,\xi\,(k\cdot x)}-\frac{i\,\zeta\,(x\cdot\gamma\mathrm{a}\gamma)\,e}{2\,m\,s\,\xi}+\frac{i\,\zeta\,(a\cdot x)\,(x\cdot\gamma\mathrm{k}\gamma)\,e}{2\,m\,s\,\xi\,(k\cdot x)}-\\
&\quad\left.\frac{D\,\zeta\,(\gamma\cdot a).(\gamma\cdot k)\,e}{2\,m\,\xi\,(k\cdot x)}+\frac{\zeta\,(\gamma\cdot a).(\gamma\cdot k)\,e}{m\,\xi\,(k\cdot x)}-\frac{\zeta\,(a\cdot\gamma\mathrm{k}\gamma)\,e}{m\,\xi\,(k\cdot x)}-\frac{\gamma\cdot x}{2\,s}-\frac{s\,\gamma\cdot k\,p^2}{k\cdot x}+\frac{\gamma\cdot k\,x^2}{4\,s\,(k\cdot x)}+\frac{i\,D\,\gamma\cdot k}{2\,(k\cdot x)}-\frac{i\,\gamma\cdot k}{k\cdot x}\right)
\end{aligned}$$

Out[53]=

$$\frac{i\,2^{-D-2}\,e^{-\frac{1}{2}\,i\,\pi\left(\frac{D}{2}-1\right)}\,\pi^{-\frac{D}{2}-1}\,\Lambda^{4-D}\,s^{-D/2}}{\mathrm{DdInv}(\zeta)}$$

Out[54]=

$$\frac{1}{12}\,a^2\,e^2\,s\,(k\cdot x)^2+e\,(a\cdot x)\,(k\cdot X)-p^2\,s-\frac{x^2}{4\,s}$$

In[55]:=
```
Matrix8 = Collect[
  Expand[
   DiracSimplify[
      Matrix7 /.
       {Pair[Momentum[x, D], Momentum[γkγ, D]] →
         DiracGamma[Momentum[k, D], D].DiracGamma[Momentum[x, D], D],
        Pair[Momentum[a, D], Momentum[γkγ, D]] →
         DiracGamma[Momentum[k, D], D].DiracGamma[Momentum[a, D], D],
        Pair[Momentum[x, D], Momentum[γaγ, D]] →
         DiracGamma[Momentum[a, D], D].DiracGamma[Momentum[x, D], D],
        Pair[Momentum[a, D], Momentum[γaγ, D]] →
         DiracGamma[Momentum[a, D], D].DiracGamma[Momentum[a, D], D]}
     ] /. FieldSubstitutions /. akToF /. {av2 → -ξ^2 m^2 / e^2}
  ],
  {sf, v1, v2, tf, af, ξ, σF}]
Coeff8 = Coeff7
Phase8 = Phase7 /. {av2 → -ξ^2 m^2 / e^2}
```

Out[55]=
$$\mathrm{v1}\left(-\frac{1}{2}\, i\, e\,(\gamma\cdot \mathrm{FDx}).\overline{\gamma}^5+\zeta\left(\frac{i\, e\,(a\cdot x)\,(\gamma\cdot k).(\gamma\cdot x)}{2\, m\, \xi\, s\,(k\cdot x)}-\frac{i\, e\,(\gamma\cdot a).(\gamma\cdot x)}{2\, m\, \xi\, s}+\sigma\mathrm{F}\left(-\frac{i\, D\, e}{4\, m\, \xi\,(k\cdot x)}-\frac{e\, p^2\, s}{2\, m\, \xi\,(k\cdot x)}+\frac{1}{8}\, e\, m\, \xi\, s\,(k\cdot x)-\frac{e\, x^2}{8\, m\, \xi\, s\,(k\cdot x)}+\frac{i\, e}{m\, \xi\,(k\cdot x)}\right)-\frac{1}{2}\, i\, m\, \xi\,(k\cdot x)\right)+\frac{i\, D\,\gamma\cdot k}{2\,(k\cdot x)}+\frac{1}{4}\, m^2\, \xi^2\, s\,\gamma\cdot k\,(k\cdot x)-\frac{p^2\, s\,\gamma\cdot k}{k\cdot x}+\frac{x^2\,\gamma\cdot k}{4\, s\,(k\cdot x)}-\frac{i\,\gamma\cdot k}{k\cdot x}-\frac{\gamma\cdot x}{2\, s}\right)+$$
$$\mathrm{af}\left(\frac{e\,(\gamma\cdot \mathrm{FDx}).\overline{\gamma}^5}{\xi\,(k\cdot x)}+\zeta\left(m+\frac{1}{2}\, i\, e\, m\, s\,\sigma\mathrm{F}\right)+i\, m^2\, \xi\, s\,\gamma\cdot k\right)+$$
$$\mathrm{sf}\left(\zeta\left(\frac{e\,(\gamma\cdot \mathrm{FDx}).\overline{\gamma}^5}{\xi\,(k\cdot x)}+i\, m^2\, \xi\, s\,\gamma\cdot k\right)+\frac{1}{2}\, i\, e\, m\, s\,\sigma\mathrm{F}+m\right)+$$
$$\mathrm{tf}\left(-\frac{2\, e\, m\, s\,\sigma\mathrm{F}}{\xi\,(k\cdot x)}-\frac{4\,\zeta\, m^2\, s\,\gamma\cdot k}{k\cdot x}\right)+$$
$$\mathrm{v2}\left(-\frac{e\,\zeta\, m\, s\,\sigma\mathrm{F}}{\xi\,(k\cdot x)}-\frac{2\, m^2\, s\,\gamma\cdot k}{k\cdot x}\right)$$

Out[56]=
$$\frac{i\, 2^{-D-2}\, e^{-\frac{1}{2}\, i\,\pi\left(\frac{D}{2}-1\right)}\,\pi^{-\frac{D}{2}-1}\,\Lambda^{4-D}\, s^{-D/2}}{\mathrm{DdInv}(\zeta)}$$

Out[57]=
$$-\frac{1}{12}\, m^2\, \xi^2\, s\,(\, k\cdot x\,)^2+e\,(a\cdot x)\,(k\cdot X)-p^2\, s-\frac{x^2}{4\, s}$$

In[58]:=
```
(*a small test*)
Expand[((Matrix8 /. {sf → 1, v1 → -1, v2 → 0, tf → 0, af → 0, ξ → 1}) +
    (Matrix8 /. {sf → 1, v1 → -1, v2 → 0, tf → 0, af → 0, ξ → -1})) / 2]
```

Out[58]=
$$\frac{1}{2}\, i\, e\,(\gamma\cdot \mathrm{FDx}).\overline{\gamma}^5-\frac{i\, D\,\gamma\cdot k}{2\,(k\cdot x)}+\frac{1}{2}\, i\, e\, m\, s\,\sigma\mathrm{F}-\frac{1}{4}\, m^2\, \xi^2\, s\,\gamma\cdot k\,(k\cdot x)+\frac{p^2\, s\,\gamma\cdot k}{k\cdot x}-\frac{x^2\,\gamma\cdot k}{4\, s\,(k\cdot x)}+\frac{i\,\gamma\cdot k}{k\cdot x}+m+\frac{\gamma\cdot x}{2\, s}$$

In[59]:=
```
DiracSimplify[
 DiracGamma[Momentum[Fx, D], D].DiracGamma[Momentum[x, D], D] /. FxToak]
DiracSimplify[
 I DiracGamma[Momentum[FDx, D], D].DiracGamma[5].DiracGamma[Momentum[x, D], D] /.
   FxToEps /. FxToak /. Gamma5toTrippleGammax]
```

Out[59]= $(a\cdot x)\,(\gamma\cdot k).(\gamma\cdot x) - (k\cdot x)\,(\gamma\cdot a).(\gamma\cdot x)$

Out[60]= $(a\cdot x)\,(\gamma\cdot k).(\gamma\cdot x) - (k\cdot x)\,(\gamma\cdot a).(\gamma\cdot x) + x^2\,(\gamma\cdot a).(\gamma\cdot k)$

In[61]:=
```
Matrix9 = Collect[
  Expand[
   Matrix8 /. {ax DiracGamma[Momentum[k, D], D].DiracGamma[Momentum[x, D], D] →
       I DiracGamma[Momentum[FDx, D], D].
          DiracGamma[5].DiracGamma[Momentum[x, D], D] +
        kx DiracGamma[Momentum[a, D], D].DiracGamma[Momentum[x, D], D] -
        ( xv2 DiracGamma[Momentum[a, D], D].DiracGamma[Momentum[k, D], D] /.
          akToF),
      DiracGamma[Momentum[k, D], D] →
       - DiracGamma[Momentum[FFx, D], D] / av2 / kx} /. {av2 → -ξ^2 m^2 / e^2}
  ],
  {sf, v1, v2, tf, af, ζ, e σF / m / ξ / kx, σF,
   e^2 DiracGamma[Momentum[FFx, D], D] / m^2 / ξ^2 / kx^2}]
Coeff9 = Coeff8
Phase9 = Phase8
```

Out[61]=
$$\mathrm{v1}\left(\zeta\left(-\frac{e\,(\gamma\cdot\mathrm{FDx}).\bar{\gamma}^5.(\gamma\cdot x)}{2\,m\,\xi\,s\,(k\cdot x)} + \frac{e\,\sigma\mathrm{F}\left(-\frac{i\,D}{4} - \frac{p^2\,s}{2} + \frac{x^2}{8\,s} + i\right)}{m\,\xi\,(k\cdot x)} + \frac{1}{8}\,e\,m\,\xi\,s\,\sigma\mathrm{F}\,(k\cdot x) - \frac{1}{2}\,i\,m\,\xi\,(k\cdot x)\right) - \frac{1}{2}\,i\,e\,(\gamma\cdot\mathrm{FDx}).\bar{\gamma}^5 + \frac{e^2\,\gamma\cdot\mathrm{FFx}\left(\frac{i\,D}{2} - p^2\,s + \frac{x^2}{4\,s} - i\right)}{m^2\,\xi^2\,(k\cdot x)^2} + \frac{1}{4}\,e^2\,s\,\gamma\cdot\mathrm{FFx} - \frac{\gamma\cdot x}{2\,s}\right) +$$
$$\mathrm{af}\left(\frac{e\,(\gamma\cdot\mathrm{FDx}).\bar{\gamma}^5}{\xi\,(k\cdot x)} + \frac{i\,e^2\,s\,\gamma\cdot\mathrm{FFx}}{\xi\,(k\cdot x)} + \zeta\left(m + \frac{1}{2}\,i\,e\,m\,s\,\sigma\mathrm{F}\right)\right) + \mathrm{sf}\left(\zeta\left(\frac{e\,(\gamma\cdot\mathrm{FDx}).\bar{\gamma}^5}{\xi\,(k\cdot x)} + \frac{i\,e^2\,s\,\gamma\cdot\mathrm{FFx}}{\xi\,(k\cdot x)}\right) + \frac{1}{2}\,i\,e\,m\,s\,\sigma\mathrm{F} + m\right) +$$
$$\mathrm{tf}\left(-\frac{4\,e^2\,\zeta\,s\,\gamma\cdot\mathrm{FFx}}{\xi^2\,(k\cdot x)^2} - \frac{2\,e\,m\,s\,\sigma\mathrm{F}}{\xi\,(k\cdot x)}\right) + \mathrm{v2}\left(-\frac{2\,e^2\,s\,\gamma\cdot\mathrm{FFx}}{\xi^2\,(k\cdot x)^2} - \frac{e\,\zeta\,m\,s\,\sigma\mathrm{F}}{\xi\,(k\cdot x)}\right)$$

Out[62]=
$$\frac{i\,2^{-D-2}\,e^{-\frac{1}{2}\,i\,\pi\left(\frac{D}{2}-1\right)}\,\pi^{-\frac{D}{2}-1}\,\Lambda^{4-D}\,s^{-D/2}}{\mathrm{DdInv}(\zeta)}$$

Out[63]=
$$-\frac{1}{12}\,m^2\,\xi^2\,s\,(k\cdot x)^2 + e\,(a\cdot x)\,(k\cdot X) - p^2\,s - \frac{x^2}{4\,s}$$

In[64]:=
```
Collect[Expand[
  Simplify[D[Coeff8 s f[χ[s]] Exp[I Phase8], s] / Exp[I Phase8] / Coeff8]], f[χ[s]]]
pv2Sol = pv2 /. Solve[% == dfds, pv2 ][[1]]
```

Out[64]=
$$s\,\chi'(s)\,f'(\chi(s)) + f(\chi(s))\left(-\frac{1}{12}\,i\,m^2\,\xi^2\,s\,(k\cdot x)^2 - \frac{D}{2} - i\,p^2\,s + \frac{i\,x^2}{4\,s} + 1\right)$$

Out[65]=
$$\frac{-m^2\,\xi^2\,s^2\,f(\chi(s))\,(k\cdot x)^2 + 6\,i\left(D\,s\,f(\chi(s)) + 2\,\mathrm{dfds}\,s - 2\,s^2\,\chi'(s)\,f'(\chi(s)) - 2\,s\,f(\chi(s))\right) + 3\,x^2\,f(\chi(s))}{12\,s^2\,f(\chi(s))}$$

Let

$f\left(\zeta,\, p^2,\, \chi\,(s)\right) = v_1\left(p^2,\, \chi\,(s)\right) D_{\zeta}^{-1}\left(p^2,\, \chi\,(s)\right)$

In[66]:=
```
Matrix9pv2 =
  Coefficient[Matrix9, pv2] /. {v1 → f[ζ, pv2, χ[s]]} /. {χ[s] → ξ kx / 2 / m^2 / s}
CoeffNoD = Coeff9 * DdInv[ζ]
PhaseNopv2 = Phase9 - Coefficient[Phase9, pv2] pv2
Collect[
  Expand[
   Simplify[-I
     D[Matrix9pv2 * CoeffNoD * Exp[I PhaseNopv2], s] / (CoeffNoD * Exp[I PhaseNopv2])
   ]
  ],
  {f[ζ, pv2, ξ kx / 2 / m^2 / s]}]
Matrix91 = Coefficient[%, f[ζ, pv2, ξ kx / 2 / m^2 / s]] v1
Matrix92 = %% - Matrix91 / v1 * f[ζ, pv2, ξ kx / 2 / m^2 / s]
```

Out[66]= $$-\frac{e^2\, s\, \gamma\cdot\text{FFx}\, f\!\left(\zeta, p^2, \frac{\xi\,(k\cdot x)}{2\,m^2\,s}\right)}{m^2\,\xi^2\,(k\cdot x)^2} - \frac{e\,\zeta\,s\,\sigma\text{F}\, f\!\left(\zeta, p^2, \frac{\xi\,(k\cdot x)}{2\,m^2\,s}\right)}{2\,m\,\xi\,(k\cdot x)}$$

Out[67]= $$i\,2^{-D-2}\, e^{-\frac{1}{2}\,i\,\pi\left(\frac{D}{2}-1\right)}\,\pi^{-\frac{D}{2}-1}\,\Lambda^{4-D}\,s^{-D/2}$$

Out[68]= $$-\frac{1}{12}\,m^2\,\xi^2\,s\,(k\cdot x)^2 + e\,(a\cdot x)\,(k\cdot X) - \frac{x^2}{4\,s}$$

Out[69]= $$f\!\left(\zeta, p^2, \frac{\xi\,(k\cdot x)}{2\,m^2\,s}\right)\left(-\frac{i\,D\,e^2\,\gamma\cdot\text{FFx}}{2\,m^2\,\xi^2\,(k\cdot x)^2} - \frac{e^2\,x^2\,\gamma\cdot\text{FFx}}{4\,m^2\,\xi^2\,s\,(k\cdot x)^2} + \frac{i\,e^2\,\gamma\cdot\text{FFx}}{m^2\,\xi^2\,(k\cdot x)^2} - \frac{i\,D\,e\,\zeta\,\sigma\text{F}}{4\,m\,\xi\,(k\cdot x)} + \frac{1}{12}\,e^2\,s\,\gamma\cdot\text{FFx} + \frac{1}{24}\,e\,\zeta\,m\,\xi\,s\,\sigma\text{F}\,(k\cdot x) - \frac{e\,\zeta\,\sigma\text{F}\,x^2}{8\,m\,\xi\,s\,(k\cdot x)} + \frac{i\,e\,\zeta\,\sigma\text{F}}{2\,m\,\xi\,(k\cdot x)}\right) - \frac{i\,e^2\,\gamma\cdot\text{FFx}\, f^{(0,0,1)}\!\left(\zeta, p^2, \frac{\xi\,(k\cdot x)}{2\,m^2\,s}\right)}{2\,m^4\,\xi\,s\,(k\cdot x)} - \frac{i\,e\,\zeta\,\sigma\text{F}\, f^{(0,0,1)}\!\left(\zeta, p^2, \frac{\xi\,(k\cdot x)}{2\,m^2\,s}\right)}{4\,m^3\,s}$$

Out[70]= $$\text{v1}\left(-\frac{i\,D\,e^2\,\gamma\cdot\text{FFx}}{2\,m^2\,\xi^2\,(k\cdot x)^2} - \frac{e^2\,x^2\,\gamma\cdot\text{FFx}}{4\,m^2\,\xi^2\,s\,(k\cdot x)^2} + \frac{i\,e^2\,\gamma\cdot\text{FFx}}{m^2\,\xi^2\,(k\cdot x)^2} - \frac{i\,D\,e\,\zeta\,\sigma\text{F}}{4\,m\,\xi\,(k\cdot x)} + \frac{1}{12}\,e^2\,s\,\gamma\cdot\text{FFx} + \frac{1}{24}\,e\,\zeta\,m\,\xi\,s\,\sigma\text{F}\,(k\cdot x) - \frac{e\,\zeta\,\sigma\text{F}\,x^2}{8\,m\,\xi\,s\,(k\cdot x)} + \frac{i\,e\,\zeta\,\sigma\text{F}}{2\,m\,\xi\,(k\cdot x)}\right)$$

Out[71]= $$-\frac{i\,e^2\,\gamma\cdot\text{FFx}\, f^{(0,0,1)}\!\left(\zeta, p^2, \frac{\xi\,(k\cdot x)}{2\,m^2\,s}\right)}{2\,m^4\,\xi\,s\,(k\cdot x)} - \frac{i\,e\,\zeta\,\sigma\text{F}\, f^{(0,0,1)}\!\left(\zeta, p^2, \frac{\xi\,(k\cdot x)}{2\,m^2\,s}\right)}{4\,m^3\,s}$$

In[72]:=
```
Matrix101 = Collect[
   (Expand[Matrix9] - Coefficient[Matrix9, pv2] pv2 + Matrix91) / m,
   {sf, v1, v2, tf, af, ξ, σF, DiracGamma[Momentum[FFx, D], D],
    DiracGamma[Momentum[x, D], D], DiracGamma[Momentum[FDx, D], D]}, Simplify]
Matrix102 = Expand[Matrix92 / m]
(*Matrix101=
 Collect[Matrix9/m,{sf,v1,v2,tf,af,ξ,σF, DiracGamma[Momentum[FFx,D],D],
     DiracGamma[Momentum[x,D],D],DiracGamma[Momentum[FDx,D],D]},Simplify]
   Matrix102=0*)
Coeff10 = Coeff9 * m
Phase10 = Phase9
```

Out[72]=
$$\mathrm{af}\left(\frac{e\,(\gamma\cdot\mathrm{FDx}).\bar{\gamma}^5}{m\,\xi\,(k\cdot x)}+\frac{i\,e^2\,s\,\gamma\cdot\mathrm{FFx}}{m\,\xi\,(k\cdot x)}+\zeta\left(1+\frac{1}{2}\,i\,e\,s\,\sigma\mathrm{F}\right)\right)+$$
$$\mathrm{v1}\left(\zeta\left(-\frac{e\,(\gamma\cdot\mathrm{FDx}).\bar{\gamma}^5.(\gamma\cdot x)}{2\,m^2\,\xi\,s\,(k\cdot x)}+\sigma\mathrm{F}\left(\frac{1}{6}\,e\,\xi\,s\,(k\cdot x)-\frac{i\,(D-3)\,e}{2\,m^2\,\xi\,(k\cdot x)}\right)-\frac{1}{2}\,i\,\xi\,(k\cdot x)\right)-\right.$$
$$\left.\frac{i\,e\,(\gamma\cdot\mathrm{FDx}).\bar{\gamma}^5}{2\,m}+\frac{e^2\,s\,\gamma\cdot\mathrm{FFx}}{3\,m}-\frac{\gamma\cdot x}{2\,m\,s}\right)+\mathrm{sf}\left(\zeta\left(\frac{e\,(\gamma\cdot\mathrm{FDx}).\bar{\gamma}^5}{m\,\xi\,(k\cdot x)}+\frac{i\,e^2\,s\,\gamma\cdot\mathrm{FFx}}{m\,\xi\,(k\cdot x)}\right)+\frac{1}{2}\,i\,e\,s\,\sigma\mathrm{F}+1\right)+$$
$$\mathrm{tf}\left(-\frac{4\,e^2\,\zeta\,s\,\gamma\cdot\mathrm{FFx}}{m\,\xi^2\,(k\cdot x)^2}-\frac{2\,e\,s\,\sigma\mathrm{F}}{\xi\,(k\cdot x)}\right)+\mathrm{v2}\left(-\frac{2\,e^2\,s\,\gamma\cdot\mathrm{FFx}}{m\,\xi^2\,(k\cdot x)^2}-\frac{e\,\zeta\,s\,\sigma\mathrm{F}}{\xi\,(k\cdot x)}\right)$$

Out[73]=
$$-\frac{i\,e^2\,\gamma\cdot\mathrm{FFx}\,f^{(0,0,1)}\!\left(\zeta,p^2,\frac{\xi\,(k\cdot x)}{2\,m^2\,s}\right)}{2\,m^5\,\xi\,s\,(k\cdot x)}-\frac{i\,e\,\zeta\,\sigma\mathrm{F}\,f^{(0,0,1)}\!\left(\zeta,p^2,\frac{\xi\,(k\cdot x)}{2\,m^2\,s}\right)}{4\,m^4\,s}$$

Out[74]=
$$\frac{i\,2^{-D-2}\,e^{-\frac{1}{2}\,i\,\pi\left(\frac{D}{2}-1\right)}\,\pi^{-\frac{D}{2}-1}\,m\,\Lambda^{4-D}\,s^{-D/2}}{\mathrm{DdInv}(\zeta)}$$

Out[75]=
$$-\frac{1}{12}\,m^2\,\xi^2\,s\,(k\cdot x)^2+e\,(a\cdot x)\,(k\cdot X)-p^2\,s-\frac{x^2}{4\,s}$$

In[76]:=
```
Matrix101 // StandardForm;
```

In[77]:=
```
Matrix10 =
  Collect[Expand[Matrix101 + Matrix102], {e^2 DiracGamma[Momentum[FFx, D], D],
    e DiracGamma[Momentum[FDx, D], D].DiracGamma[5],
    e DiracGamma[Momentum[FDx, D], D].DiracGamma[5].DiracGamma[Momentum[x, D], D],
    σF, DiracGamma[Momentum[x, D], D], sf, v1, v2, tf, af, ξ}]
```

Out[77]=
$$e\,(\gamma\cdot\mathrm{FDx}).\bar{\gamma}^5\left(\frac{\mathrm{af}}{m\,\xi\,(k\cdot x)}+\frac{\zeta\,\mathrm{sf}}{m\,\xi\,(k\cdot x)}-\frac{i\,\mathrm{v1}}{2\,m}\right)-\frac{e\,\zeta\,\mathrm{v1}\,(\gamma\cdot\mathrm{FDx}).\bar{\gamma}^5.(\gamma\cdot x)}{2\,m^2\,\xi\,s\,(k\cdot x)}+$$
$$e^2\,\gamma\cdot\mathrm{FFx}\left(-\frac{i\,f^{(0,0,1)}\!\left(\zeta,p^2,\frac{\xi\,(k\cdot x)}{2\,m^2\,s}\right)}{2\,m^5\,\xi\,s\,(k\cdot x)}-\frac{4\,\zeta\,s\,\mathrm{tf}}{m\,\xi^2\,(k\cdot x)^2}-\frac{2\,s\,\mathrm{v2}}{m\,\xi^2\,(k\cdot x)^2}+\frac{i\,\mathrm{af}\,s}{m\,\xi\,(k\cdot x)}+\frac{i\,\zeta\,s\,\mathrm{sf}}{m\,\xi\,(k\cdot x)}+\frac{s\,\mathrm{v1}}{3\,m}\right)+$$
$$\sigma\mathrm{F}\left(-\frac{i\,e\,\zeta\,f^{(0,0,1)}\!\left(\zeta,p^2,\frac{\xi\,(k\cdot x)}{2\,m^2\,s}\right)}{4\,m^4\,s}+\frac{1}{2}\,i\,\mathrm{af}\,e\,\zeta\,s+\zeta\,\mathrm{v1}\left(-\frac{i\,D\,e}{2\,m^2\,\xi\,(k\cdot x)}+\frac{3\,i\,e}{2\,m^2\,\xi\,(k\cdot x)}+\frac{1}{6}\,e\,\xi\,s\,(k\cdot x)\right)-\right.$$
$$\left.\frac{2\,e\,s\,\mathrm{tf}}{\xi\,(k\cdot x)}-\frac{e\,\zeta\,s\,\mathrm{v2}}{\xi\,(k\cdot x)}+\frac{1}{2}\,i\,e\,s\,\mathrm{sf}\right)+\mathrm{af}\,\zeta-\frac{1}{2}\,i\,\zeta\,\xi\,\mathrm{v1}\,(k\cdot x)-\frac{\mathrm{v1}\,\gamma\cdot x}{2\,m\,s}+\mathrm{sf}$$

In[78]:=
```
Matrix10ReorderPart1 =
  Collect[Coefficient[Matrix10, e^2 DiracGamma[Momentum[FFx, D], D]] *
      m^1 ξ^2 (kx)^2, {sf, v1, v2, tf, af, ξ}, Simplify]
    e^2 DiracGamma[Momentum[FFx, D], D] / (m^1 ξ^2 (kx)^2) +
   Collect[
     Coefficient[Matrix10, e DiracGamma[Momentum[FDx, D], D].DiracGamma[5]] *
      m ξ (kx), {sf, v1, v2, tf, af, ξ}, Simplify] e
    DiracGamma[Momentum[FDx, D], D].DiracGamma[5] / (m ξ (kx)) +
   Collect[Coefficient[Matrix10, e σF] * 2 m^0 ξ (kx) / ξ,
     {sf, v1, v2, tf, af, ξ}, Simplify] e σF / (2 m^0 ξ (kx) / ξ) +
   Coefficient[Matrix10, e DiracGamma[Momentum[FDx, D], D].DiracGamma[5].
       DiracGamma[Momentum[x, D], D]] e DiracGamma[Momentum[FDx, D], D].
     DiracGamma[5].DiracGamma[Momentum[x, D], D] +
   Coefficient[Matrix10, DiracGamma[Momentum[x, D], D]] ×
    DiracGamma[Momentum[x, D], D];
Matrix10Reorder =
 Matrix10ReorderPart1 + Expand[Matrix10 - Matrix10ReorderPart1] /. {1 / ξ → ξ}
Coeff10
Phase10
```

Out[79]=
$$\frac{e\,(\gamma\cdot\text{FDx}).\bar{\gamma}^5\left(\text{af}-\frac{1}{2}\,i\,\xi\,\text{v1}\,(k\cdot x)+\zeta\,\text{sf}\right)}{m\,\xi\,(k\cdot x)}-\frac{e\,\zeta\,\text{v1}\,(\gamma\cdot\text{FDx}).\bar{\gamma}^5.(\gamma\cdot x)}{2\,m^2\,\xi\,s\,(k\cdot x)}+$$
$$\frac{e\,\zeta\,\sigma\text{F}\left(-\frac{i\,\xi\,(k\cdot x)\,f^{(0,0,1)}\left(\zeta,p^2,\frac{\xi\,(k\cdot x)}{2\,m^2\,s}\right)}{2\,m^4\,s}+\frac{\text{v1}\left(m^2\,\xi^2\,s\,(k\cdot x)^2-3\,i\,D+9\,i\right)}{3\,m^2}+i\,\text{af}\,\xi\,s\,(k\cdot x)+i\,\zeta\,\xi\,s\,\text{sf}\,(k\cdot x)-4\,\zeta\,s\,\text{tf}-2\,s\,\text{v2}\right)}{2\,\xi\,(k\cdot x)}+$$
$$\frac{e^2\,\gamma\cdot\text{FFx}\left(-\frac{i\,\xi\,(k\cdot x)\,f^{(0,0,1)}\left(\zeta,p^2,\frac{\xi\,(k\cdot x)}{2\,m^2\,s}\right)}{2\,m^4\,s}+\frac{1}{3}\,\xi^2\,s\,\text{v1}\,(k\cdot x)^2+i\,\text{af}\,\xi\,s\,(k\cdot x)+i\,\zeta\,\xi\,s\,\text{sf}\,(k\cdot x)-4\,\zeta\,s\,\text{tf}-2\,s\,\text{v2}\right)}{m\,\xi^2\,(k\cdot x)^2}+$$
$$\text{af}\,\zeta-\frac{1}{2}\,i\,\zeta\,\xi\,\text{v1}\,(k\cdot x)-\frac{\text{v1}\,\gamma\cdot x}{2\,m\,s}+\text{sf}$$

Out[80]=
$$\frac{i\,2^{-D-2}\,e^{-\frac{1}{2}\,i\,\pi\left(\frac{D}{2}-1\right)}\,\pi^{-\frac{D}{2}-1}\,m\,\Lambda^{4-D}\,s^{-D/2}}{\text{DdInv}(\zeta)}$$

Out[81]=
$$-\frac{1}{12}\,m^2\,\xi^2\,s\,(k\cdot x)^2+e\,(a\cdot x)\,(k\cdot X)-p^2\,s-\frac{x^2}{4\,s}$$

Matrix10 without removing p^2 with integration by parts

In[82]:=
```
sf + af ζ - (v1 DiracGamma[Momentum[x, D], D])/(2 m s) -
  (e v1 ζ DiracGamma[Momentum[FDx, D], D].DiracGamma[5].
      DiracGamma[Momentum[x, D], D]) /
   (2 m^2 s ξ Pair[Momentum[k, D], Momentum[x, D]]) -
  1/2 ⅈ v1 ζ ξ Pair[Momentum[k, D], Momentum[x, D]] +
  (e DiracGamma[Momentum[FDx, D], D].DiracGamma[5]
      (af + sf ζ - 1/2 ⅈ v1 ξ Pair[Momentum[k, D], Momentum[x, D]])) /
   (m ξ Pair[Momentum[k, D], Momentum[x, D]]) +
  (e ζ σF (-2 s v2 - 4 s tf ζ + ⅈ af s ξ Pair[Momentum[k, D], Momentum[x, D]] +
        ⅈ s sf ζ ξ Pair[Momentum[k, D], Momentum[x, D]] + 1/(4 m^2 s) v1 (8 ⅈ s - 2 ⅈ D s +
           m^2 s^2 ξ^2 Pair[Momentum[k, D], Momentum[x, D]]^2 - 4 s^2 Pair[Momentum[p, D],
              Momentum[p, D]] + Pair[Momentum[x, D], Momentum[x, D]]))) /
   (2 ξ Pair[Momentum[k, D], Momentum[x, D]]) + (e^2 DiracGamma[Momentum[FFx, D], D]
     (-2 s v2 - 4 s tf ζ + ⅈ af s ξ Pair[Momentum[k, D], Momentum[x, D]] +
       ⅈ s sf ζ ξ Pair[Momentum[k, D], Momentum[x, D]] +
       1/(4 m^2 s) v1 (-4 ⅈ s + 2 ⅈ D s + m^2 s^2 ξ^2 Pair[Momentum[k, D], Momentum[x, D]]^2 -
          4 s^2 Pair[Momentum[p, D], Momentum[p, D]] + Pair[Momentum[x, D],
            Momentum[x, D]]))) / (m ξ^2 Pair[Momentum[k, D], Momentum[x, D]]^2)
```

Out[82]=

$$\frac{e\,(\gamma\cdot \mathrm{FDx}).\bar{\gamma}^5\left(\mathrm{af}-\frac{1}{2}\,i\,\xi\,\mathrm{v1}\,(k\cdot x)+\zeta\,\mathrm{sf}\right)}{m\,\xi\,(k\cdot x)}-\frac{e\,\zeta\,\mathrm{v1}\,(\gamma\cdot \mathrm{FDx}).\bar{\gamma}^5.(\gamma\cdot x)}{2\,m^2\,\xi\,s\,(k\cdot x)}+$$

$$\frac{e^2\,\gamma\cdot \mathrm{FFx}\left(\frac{\mathrm{v1}\left(m^2\,\xi^2\,s^2\,(k\cdot x)^2+2\,i\,D\,s-4\,p^2\,s^2-4\,i\,s+x^2\right)}{4\,m^2\,s}+i\,\mathrm{af}\,\xi\,s\,(k\cdot x)+i\,\zeta\,\xi\,s\,\mathrm{sf}\,(k\cdot x)-4\,\zeta\,s\,\mathrm{tf}-2\,s\,\mathrm{v2}\right)}{m\,\xi^2\,(k\cdot x)^2}+$$

$$\frac{e\,\zeta\,\sigma\mathrm{F}\left(\frac{\mathrm{v1}\left(m^2\,\xi^2\,s^2\,(k\cdot x)^2-2\,i\,D\,s-4\,p^2\,s^2+8\,i\,s+x^2\right)}{4\,m^2\,s}+i\,\mathrm{af}\,\xi\,s\,(k\cdot x)+i\,\zeta\,\xi\,s\,\mathrm{sf}\,(k\cdot x)-4\,\zeta\,s\,\mathrm{tf}-2\,s\,\mathrm{v2}\right)}{2\,\xi\,(k\cdot x)}+$$

$$\mathrm{af}\,\zeta-\frac{1}{2}\,i\,\zeta\,\xi\,\mathrm{v1}\,(k\cdot x)-\frac{\mathrm{v1}\,\gamma\cdot x}{2\,m\,s}+\mathrm{sf}$$

In[83]:=
```
(*a small test*)
Expand[((Matrix101 /. {sf → 1, v1 → -1, v2 → 0, tf → 0, af → 0, ζ → 1}) +
    (Matrix101 /. {sf → 1, v1 → -1, v2 → 0, tf → 0, af → 0, ζ → -1})) / 2]
```

Out[83]=

$$\frac{i\,e\,(\gamma\cdot \mathrm{FDx}).\bar{\gamma}^5}{2\,m}-\frac{e^2\,s\,\gamma\cdot \mathrm{FFx}}{3\,m}+\frac{1}{2}\,i\,e\,s\,\sigma\mathrm{F}+\frac{\gamma\cdot x}{2\,m\,s}+1$$

In[84]:=
```
Matrix111 = Collect[
  DiracOrder[DiracSimplify[Expand[Matrix101] /. FxToEps /. FxToak /.
     Gamma5toTrippleGammax /. {av2 → -ξ^2 m^2 / e^2}]],
  {sf, v1, v2, tf, af, ζ}, Simplify]
Matrix112 = Expand[Matrix102 /. FxToEps /. FxToak /. Gamma5toTrippleGammax]
Coeff10
Phase10
```

Out[84]=
$$\mathrm{v1}\left(\frac{s\left(\gamma\cdot k\left(2\,m^2\,\xi^2\,s\,(k\cdot x)-3\,e\,(a\cdot x)\right)+3\,e\,\gamma\cdot a\,(k\cdot x)-3\,e\,(\gamma\cdot a).(\gamma\cdot k).(\gamma\cdot x)\right)-3\,\gamma\cdot x}{6\,m\,s}-\frac{1}{6\,m^2\,\xi\,s\,(k\cdot x)}i\,\zeta\left(e\,(\gamma\cdot a).(\gamma\cdot k)\left(2\,m^2\,\xi^2\,s^2\,(k\cdot x)^2-6\,i\,(D-3)\,s-3\,x^2\right)+3\,(k\cdot x)\left(e\,(\gamma\cdot a).(\gamma\cdot x)+m^2\,\xi^2\,s\,(k\cdot x)\right)-3\,e\,(a\cdot x)\,(\gamma\cdot k).(\gamma\cdot x)\right)\right)+$$
$$\mathrm{af}\left(\zeta\,(e\,s\,(\gamma\cdot a).(\gamma\cdot k)+1)+\frac{i\left(\gamma\cdot k\left(m^2\,\xi^2\,s\,(k\cdot x)-e\,(a\cdot x)\right)+e\,\gamma\cdot a\,(k\cdot x)-e\,(\gamma\cdot a).(\gamma\cdot k).(\gamma\cdot x)\right)}{m\,\xi\,(k\cdot x)}\right)+$$
$$\mathrm{sf}\left(\frac{i\,\zeta\left(\gamma\cdot k\left(m^2\,\xi^2\,s\,(k\cdot x)-e\,(a\cdot x)\right)+e\,\gamma\cdot a\,(k\cdot x)-e\,(\gamma\cdot a).(\gamma\cdot k).(\gamma\cdot x)\right)}{m\,\xi\,(k\cdot x)}+e\,s\,(\gamma\cdot a).(\gamma\cdot k)+1\right)+$$
$$\mathrm{tf}\left(-\frac{4\,\zeta\,m\,s\,\gamma\cdot k}{k\cdot x}+\frac{4\,i\,e\,s\,(\gamma\cdot a).(\gamma\cdot k)}{\xi\,(k\cdot x)}\right)+$$
$$\mathrm{v2}\left(-\frac{2\,m\,s\,\gamma\cdot k}{k\cdot x}+\frac{2\,i\,e\,\zeta\,s\,(\gamma\cdot a).(\gamma\cdot k)}{\xi\,(k\cdot x)}\right)$$

Out[85]=
$$-\frac{e\,\zeta\,(\gamma\cdot a).(\gamma\cdot k)\,f^{(0,0,1)}\left(\zeta,p^2,\frac{\xi\,(k\cdot x)}{2\,m^2\,s}\right)}{2\,m^4\,s}+\frac{i\,a^2\,e^2\,\gamma\cdot k\,f^{(0,0,1)}\left(\zeta,p^2,\frac{\xi\,(k\cdot x)}{2\,m^2\,s}\right)}{2\,m^5\,\xi\,s}$$

Out[86]=
$$\frac{i\,2^{-D-2}\,e^{-\frac{1}{2}\,i\,\pi\left(\frac{D}{2}-1\right)}\,\pi^{-\frac{D}{2}-1}\,m\,\Lambda^{4-D}\,s^{-D/2}}{\mathrm{DdInv}(\zeta)}$$

Out[87]=
$$-\frac{1}{12}\,m^2\,\xi^2\,s\,(k\cdot x)^2+e\,(a\cdot x)\,(k\cdot X)-p^2\,s-\frac{x^2}{4\,s}$$

In[88]:=
```
Collect[Expand[Matrix111],
 {
  DiracGamma[Momentum[x, D], D],
  DiracGamma[Momentum[k, D], D],
  DiracGamma[Momentum[a, D], D],
  DiracGamma[Momentum[a, D], D].DiracGamma[Momentum[k, D], D] ,
  DiracGamma[Momentum[a, D], D].DiracGamma[Momentum[x, D], D] ,
  DiracGamma[Momentum[k, D], D].DiracGamma[Momentum[x, D], D] ,
  DiracGamma[Momentum[a, D], D].
   DiracGamma[Momentum[k, D], D].DiracGamma[Momentum[x, D], D],
  sf, v1, v2, tf, af, ξ}]
```

Out[88]=

$$(\gamma\cdot a).(\gamma\cdot k)\left(\text{af}\,e\,\zeta\,s+\zeta\,\text{v1}\left(-\frac{D\,e}{m^2\,\xi\,(k\cdot x)}+\frac{i\,e\,x^2}{2\,m^2\,\xi\,s\,(k\cdot x)}+\frac{3\,e}{m^2\,\xi\,(k\cdot x)}-\frac{1}{3}\,i\,e\,\xi\,s\,(k\cdot x)\right)+\frac{4\,i\,e\,s\,\text{tf}}{\xi\,(k\cdot x)}+\frac{2\,i\,e\,\zeta\,s\,\text{v2}}{\xi\,(k\cdot x)}+e\,s\,\text{sf}\right)+$$
$$\gamma\cdot k\left(\text{af}\left(i\,m\,\xi\,s-\frac{i\,e\,(a\cdot x)}{m\,\xi\,(k\cdot x)}\right)+\zeta\,\text{sf}\left(i\,m\,\xi\,s-\frac{i\,e\,(a\cdot x)}{m\,\xi\,(k\cdot x)}\right)+\text{v1}\left(\frac{1}{3}\,m\,\xi^2\,s\,(k\cdot x)-\frac{e\,(a\cdot x)}{2\,m}\right)-\frac{4\,\zeta\,m\,s\,\text{tf}}{k\cdot x}-\frac{2\,m\,s\,\text{v2}}{k\cdot x}\right)+$$
$$(\gamma\cdot a).(\gamma\cdot k).(\gamma\cdot x)\left(-\frac{i\,\text{af}\,e}{m\,\xi\,(k\cdot x)}-\frac{i\,e\,\zeta\,\text{sf}}{m\,\xi\,(k\cdot x)}-\frac{e\,\text{v1}}{2\,m}\right)+\gamma\cdot a\left(\frac{i\,\text{af}\,e}{m\,\xi}+\frac{e\,\text{v1}\,(k\cdot x)}{2\,m}+\frac{i\,e\,\zeta\,\text{sf}}{m\,\xi}\right)+$$
$$\frac{i\,e\,\zeta\,\text{v1}\,(a\cdot x)\,(\gamma\cdot k).(\gamma\cdot x)}{2\,m^2\,\xi\,s\,(k\cdot x)}-\frac{i\,e\,\zeta\,\text{v1}\,(\gamma\cdot a).(\gamma\cdot x)}{2\,m^2\,\xi\,s}+\text{af}\,\zeta-\frac{1}{2}\,i\,\zeta\,\xi\,\text{v1}\,(k\cdot x)-\frac{\text{v1}\,\gamma\cdot x}{2\,m\,s}+\text{sf}$$

Final result for the electron propagator in a CCF

$$S^c\,(x_2,\,x_1)\;=\Lambda^{4-D}\int\frac{\mathbb{d}^D p}{(2\,\pi)^D}\;E_p\,(x_2)$$
$$\sum_{\zeta=+-}\frac{\mathbb{i}}{2}\left[m\,s\,\left(p^2,\,\chi_p\right)-(\gamma p)\;v_1\,\left(p^2,\,\chi_p\right)-\frac{e^2\,\left(\gamma F^2\,p\right)}{m^4}\;v_2\,\left(p^2,\,\chi_p\right)-\frac{e\,\sigma F}{m}\;t\right.$$
$$\left.\left(p^2,\,\chi_p\right)+\frac{e\,(\gamma F^*\,p)\,\gamma^5}{m^2}\;a_s\,\left(p^2,\,\chi_p\right)\right]\frac{1+\zeta\,\gamma\epsilon^{(2)}\,\gamma^5}{D_\zeta\,\left(p^2,\,\chi_p\right)}\;E_p^{bar}\,(x_1)\;=$$
$$=\mathbb{e}^{-\mathbb{i}\,\frac{\pi}{2}\,\left(\frac{D}{2}-2\right)}\;\frac{m\,\Lambda^{4-D}}{2^{D+2}\,\pi^{D/2+1}}\;\mathbb{e}^{\mathbb{i}\,\eta}$$
$$\sum_{\zeta=+-}\int_0^\infty\frac{\mathbb{d}s}{s^{D/2}}\left(\int_0^\infty\frac{\mathbb{d}p^2}{D_\zeta\,\left(p^2,\,\chi_p\,(s)\right)}\;\mathbb{e}^{-\mathbb{i}sp^2-\mathbb{i}\,\frac{x^2}{4\,s}+\mathbb{i}\,\frac{s}{12}\,e^2\,(Fx)^2}\times\right.$$
$$\left\{s\,\left(p^2,\,\chi_p\,(s)\right)\right[$$
$$\left.1+\frac{1}{2}\;\mathbb{i}\,s\,e\,\sigma F+\zeta\left(\frac{\mathbb{i}\,e^2\,s\,(\gamma\,FFx)}{m\,\xi\,(kx)}+\frac{e\,F_{\alpha\beta}{}^*\,x^\beta\,\gamma^\alpha\,\gamma^5}{m\,\xi\,(kx)}\right)\right]$$
$$+\,v_1\,\left(p^2,\,\chi_p\,(s)\right)\left[-\frac{(\gamma x)}{2\,m\,s}+\frac{e^2\,s\,(\gamma\,FFx)}{3\,m}-\frac{\mathbb{i}\,e\,F_{\alpha\beta}{}^*\,x^\beta\,\gamma^\alpha\,\gamma^5}{2\,m}\right.$$

$$+\zeta\left(-\frac{1}{2}\,\mathbb{i}\,\xi\,(kx)-\frac{e\,\sigma F}{2\,m^2\,\xi\,(kx)}\right.$$

$$\left.\left(\mathbb{i}\,(D-3)-\frac{1}{3}\,m^2\,s\,\xi^2\,(kx)^2\right)-\frac{e\,(\gamma F^*\,x)\,\gamma^5\,(\gamma x)}{2\,m^2\,\xi\,s\,(kx)}\right)\Bigg]$$

$$+v_2\left(p^2,\chi_p\,(s)\right)\left[-\frac{2\,s\,e^2\,(\gamma\,FFx)}{m\,\xi^2\,(kx)^2}-\zeta\,\frac{s\,e\,\sigma F}{\xi\,(kx)}\right]$$

$$+t\left(p^2,\chi_p\,(s)\right)\left[-\frac{2\,s\,e\,\sigma F}{\xi\,(kx)}-\zeta\,\frac{4\,s\,e^2\,(\gamma\,FFx)}{m\,\xi^2\,(kx)^2}\right]$$

$$+a\left(p^2,\chi_p\,(s)\right)\Bigg[$$

$$\frac{e\,F_{\alpha\beta}{}^*\,x^\beta\,\gamma^\alpha\,\gamma^5}{m\,\xi\,(kx)}+\frac{\mathbb{i}\,s\,e^2\,(\gamma\,FFx)}{m\,\xi\,(kx)}+\zeta\left(1+\frac{1}{2}\,\mathbb{i}\,s\,e\,\sigma F\right)\Bigg]\Bigg\}$$

$$+\int_0^\infty\mathbb{d}p^2\,\frac{\partial}{\partial\chi_p}\left(\frac{v_1\left(p^2,\chi_p\right)}{D_\zeta\left(p^2,\chi_p\right)}\right)\,\mathbb{e}^{-\mathbb{i}sp^2-\mathbb{i}\,\frac{x^2}{4s}+\mathbb{i}\,\frac{s}{12}\,e^2\,(Fx)^2}\left[-\frac{\mathbb{i}\,e^2\,(\gamma\,FFx)}{2\,m^5\,s\,\xi\,(kx)}-\zeta\,\frac{\mathbb{i}\,e\,\sigma F}{4\,m^4\,s}\right]\Bigg)$$

$$=\mathbb{e}^{-\mathbb{i}\,\frac{\pi}{2}\left(\frac{D}{2}-2\right)}\,\frac{m\,\Lambda^{4-D}}{2^{D+2}\,\pi^{D/2+1}}\,\mathbb{e}^{\mathbb{i}\,\eta}\sum_{\zeta=+-}\int_0^\infty\frac{\mathbb{d}s}{s^{D/2}}\int_0^\infty\frac{\mathbb{d}p^2}{D_\zeta\left(p^2,\chi_p\,(s)\right)}\,\mathbb{e}^{-\mathbb{i}sp^2-\mathbb{i}\,\frac{x^2}{4s}+\mathbb{i}\,\frac{s}{12}\,e^2\,(Fx)^2}\Bigg\{$$

$$s\,(p)+\zeta\left[a\,(p)-\frac{\mathbb{i}}{2}\,\xi\,(kx)\,v_1\,(p)\right]$$

$$-\frac{(\gamma x)}{2\,m\,s}\,v_1\,(p)$$

$$+\frac{e^2\left(\gamma F^2\,x\right)}{m\,\xi^2\,(kx)^2}\Bigg[$$

$$\mathbb{i}\,\zeta\,\xi\,s\,s\,(p)+\left(\frac{1}{4}\,\xi^2\,s\,(kx)^2-s\,\frac{p^2}{m^2}+\frac{x^2}{4\,m^2\,s}+\frac{\mathbb{i}\,(D-2)}{2\,m^2}\right)v_1\,(p)-$$

$$2\,s\,v_2\,(p)-4\,\zeta\,s\,t\,(p)+\mathbb{i}\,\xi\,s\,(kx)\,a\,(p)\Bigg]$$

$$+$$

$$\zeta\,\frac{e\,\sigma F}{\xi\,(kx)}\Bigg[\mathbb{i}\,\zeta\,\xi\,s\,s\,(p)+\left(\frac{1}{4}\,\xi^2\,s\,(kx)^2-s\,\frac{p^2}{m^2}+\frac{x^2}{4\,m^2\,s}-\frac{\mathbb{i}\,(D-4)}{2\,m^2}\right)$$

$$v_1\,(p)-2\,s\,v_2\,(p)-4\,\zeta\,s\,t\,(p)+\mathbb{i}\,\xi\,s\,(kx)\,a\,(p)\Bigg]$$

$$+\frac{e\,(\gamma F^*\,x)\,\gamma^5}{m\,\xi\,(kx)}\left[\zeta\,s\,(p)-\frac{\mathbb{i}}{2}\,\xi\,(kx)\,v_1\,(p)+a\,(p)\right]$$

$$-\zeta\,\frac{e\,(\gamma F^*\,x)\,\gamma^5\,(\gamma x)}{2\,m^2\,s\,\xi\,(kx)}\,v_1\,(p)\Bigg\}$$

$$=$$

$$\mathbb{e}^{-\mathbb{i}\,\frac{\pi}{2}\left(\frac{D}{2}-2\right)}\,\frac{m\,\Lambda^{4-D}}{2^{D+2}\,\pi^{D/2+1}}\,\mathbb{e}^{\mathbb{i}\,\eta}\sum_{\zeta=+-}\int_0^\infty\frac{\mathbb{d}s}{s^{D/2}}\int_0^\infty\frac{\mathbb{d}p^2}{D_\zeta\left(p^2,\chi_p\,(s)\right)}\,\mathbb{e}^{-\mathbb{i}sp^2-\mathbb{i}\,\frac{x^2}{4s}+\mathbb{i}\,\frac{s}{12}\,e^2\,(Fx)^2}\Bigg\{$$

$$s(p) + \zeta\left[a(p) - \frac{i}{2}\xi(kx)\, v_1(p)\right]$$

$$-\frac{(\gamma x)}{2ms} v_1(p)$$

$$+\frac{e^2\left(\gamma F^2 x\right)}{m^3 \xi^2 (kx)^2} m^2 \Big[$$

$$i\zeta\xi(kx)\, s\, s(p) + \frac{1}{3} s\,\xi^2 (kx)^2 v_1(p) - 2 s\, v_2(p) - 4\zeta s\, t(p) +$$

$$i\xi s(kx)\, a(p) - \frac{i\xi(kx)}{2m^4 s} D_\zeta\left(p^2, \chi_p(s)\right) \frac{\partial}{\partial\chi_p}\left(\frac{v_1\left(p^2, \chi_p\right)}{D_\zeta\left(p^2, \chi_p\right)}\right)\Big]$$

$$+\zeta\frac{e\,\sigma F}{m^2 \xi(kx)} m^2 \Big[i\zeta\xi(kx)\, s\, s(p) +$$

$$\left(\frac{1}{3} s\,\xi^2 (kx)^2 - \frac{i(D-3)}{m^2}\right) v_1(p) - 2 s\, v_2(p) - 4\zeta s\, t(p) +$$

$$i\xi s(kx)\, a(p) - \frac{i\xi(kx)}{2m^4 s} D_\zeta\left(p^2, \chi_p(s)\right) \frac{\partial}{\partial\chi_p}\left(\frac{v_1\left(p^2, \chi_p\right)}{D_\zeta\left(p^2, \chi_p\right)}\right)\Big]$$

$$+\frac{e(\gamma F^* x)\gamma^5}{m\xi(kx)}\left[\zeta s(p) - \frac{i}{2}\xi(kx)\, v_1(p) + a(p)\right]$$

$$-\zeta\frac{e(\gamma F^* x)\gamma^5(\gamma x)}{2m^2 s\,\xi(kx)} v_1(p)\Big\}$$

$$= e^{-i\frac{\pi}{2}\left(\frac{D}{2}-1\right)} \frac{\Lambda^{4-D}}{2^D \pi^{D/2}} m^{D-1} e^{i\eta}$$

$$\int_0^\infty \frac{ds}{s^{D/2}}\left[1 + \frac{m(\gamma x)}{2s} + \frac{e(\gamma a)(kx)}{2m} - \frac{e(\gamma k)(ax)}{2m} + \frac{e(\gamma x)(\gamma a)(\gamma k)}{2m} +\right.$$

$$\left.\frac{e s(\gamma a)(\gamma k)}{m^2} + \frac{e^2 a^2 s(\gamma k)(kx)}{3m^3}\right] e^{-is - i\frac{m^2 x^2}{4s} + i\frac{s}{12}\frac{e^2}{m^2}(Fx)^2} =$$

$$= e^{-i\frac{\pi}{2}\left(\frac{D}{2}-1\right)} \frac{\Lambda^{4-D}}{2^{D+1} \pi^{D/2}} m^D e^{i\eta}$$

$$\int_0^\infty \frac{ds}{s^{D/2+1}}\left[\frac{2s}{m} + \gamma^\alpha\left(g_{\alpha\beta} - \frac{es}{m^2} F_{\alpha\beta} + \frac{e^2 s^2}{3m^4} F_{\alpha\lambda} F^\lambda{}_\beta\right) x^\beta\right]$$

$$\left(1 + \frac{ies}{2}\sigma^{\alpha\beta} F_{\alpha\beta}\right) e^{-is - i\frac{m^2 x^2}{4s} + i\frac{s}{12}\frac{e^2}{m^2}(Fx)^2}$$

$\eta = e(ax)(k, (x_1 + x_2)/2)$,

$x = x_2 - x_1$,

$e > 0$,

$\sigma^{\alpha\beta} = \frac{i}{2}\left(\gamma^\alpha\gamma^\beta - \gamma^\beta\gamma^\alpha\right)$,

$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$,

$$e^{-i\frac{\pi}{2}\left(\frac{D}{2}-1\right)}\frac{\Lambda^{4-D}}{2^D\pi^{D/2}}m^{D-1}\rightarrow\frac{(-i)\,m^3}{16\,\pi^2},\quad D\rightarrow 4$$